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DESIGN AND IMPLEMENTATION OF SOLAR TRACKING SYSTEM FOR PHOTOVOLTAIC CELLS

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مُلخص

تصميم وتنفيذ نظام خلايا كهروضوئية متتبع للشمس

في الوقت الحاضر توليد الكهرباء من الطاقة الشمسية غير كفؤ ومكلف. الطاقة المستخرجة من الألواح الشمسية تعتمد على مقدار الأشعة الشمسية الساقطة عليها ، ولاستخراج أقصى قدر من الطاقة الشمسية، يجب أن تكون الأشعة الشمسية الساقطة عمودية على اللوح. في مشروع التخرج هذا ، نسعى لتحسين كفاءة الأنظمة الشمسية من خلال تصميم وتنفيذ نظام تتبّع الأشعة الشمسية. هذا النظام سوف يبقي الألواح الشمسية تتحرك مع الشمس من أجل زيادة مقدار الطاقة الشمسية المستخرجة .

آلية عمل هذا النظام أنّه يتتّبّع الضوء الأكثر كثافة عن طريق تحريك الألواح لتكون عموديّة على الضوء الساقط. يستخدم هذا النظام اثنان من الحساسات الضوئية التي تتحدّس ألمع نقطة في السماء. تتم معالجة البيانات والإشارات القادمة من الحساسات الضوئية عن طريق متحكم، هذا المتحكم يتحكم بحركة المحرّك. تم تقييم أداء هذا النظام عن طريق تقييم المعايير الهامة مثل كمّية الإشعاع الشمسي الساقط على اللوح الشمسي ، الحد الأقصى من الطاقة الكهربائية كل ساعة ، الزيادة في الكفاءة ، تيّار الدائرة المغلقة ، جهد الدائرة المفتوحة. كما تم مقارنة المعايير السابقة بمعايير الألواح الشمسية الثابتة.

Table of Contents

Table of Contents	VI
List of Figures	VIII
List of Tables	X
Nomenclature	XI
Abstract	XII
Ch.1: INTRODUCTION	1
1.1 Overview	2
1.2 Problem Statement	3
1.3 Objectives	4
1.4 Report Organization.....	5
Ch.2: LITERATURE REVIEW	6
2.1 Overview	7
2.2 Photovoltaic Systems	7
2.3 Working of PV Cell	9
2.4 PV Mathematical Background.....	10
2.5 Previous Solar Tracker.....	13
Ch.3: PROJECT HARDWARE	15
3.1 Monocrystalline Solar panel	16
3.2 Light Dependent Resistors	17
3.3 Microcontrollers.....	19
3.4 Relay Module.....	21
3.5 DC Motor	22
3.7 Battery	23
Ch.4: METHODOLOGY	24
4.1 Overview	25
4.2 Project Planning	26
4.3 Design of Framework	30
4.4 Design of Driving Motor	32
4.5 Controlling Mechanism	33
4.6 Interfacing & Circuit Connection	37

Ch.5: SOFTWARE	38
5.1 Arduino Programming Platform	39
5.2 Tracking Algorithm	40
5.3 Software Implementation	41
Ch.6: EXPERIMENTAL VALIDATION	42
6.1 Overview	43
6.2 Experiment Results	45
6.3 Results Analysis	46
Ch.7: CONCLUSION	50
7.1 Conclusion	51
7.2 Recommendation & Future Work	52
References	53
Appendices	55
I. Programming Code	55
II. Arduino Uno Pins	66

List of Figures

Figure 1.1: Earth-sun belt area with high potential solar energy	2
Figure 1.2: Monthly temperature data for Buraydah City	4
Figure 2.1: Schematic diagram of the composition of a photovoltaic cell	8
Figure 2.2: Typical solar panel module with 36 cells connected in series	8
Figure 2.3: Photovoltaic Array hierarchy	9
Figure 2.4: Diagram of the working principle of a PV cell	10
Figure 2.5: Solar cell diagram in a short circuit and loaded circuit	11
Figure 2.6: Characteristic diagram of a solar cell.....	12
Figure 3.1: The solar cell panel used in this project.....	16
Figure 3.2: Solar cell parameters.....	16
Figure 3.3: (12mm CdS) light sensors.....	17
Figure 3.4: Dimensions and components of one LDR	17
Figure 3.5: Sensitivity & temperature characteristics of LDR	18
Figure 3.6: Arduino Uno and its external power regulator.....	19
Figure 3.7: Arduino Uno components	20
Figure 3.8: Relay module to be connected to a DC motor	21
Figure 3.9: Wiring and board setup of the relay module.....	21
Figure 3.10: Used windshield wiper DC motor driving solar panel.....	22
Figure 3.11: Windshield wiper DC motor drawings with dimensions	22
Figure 3.12: Battery powering the DC motor.....	23
Figure 4.1: Engineering design process used in the design and implementation of this tracker....	25
Figure 4.2: Project flow chart of the solar tracking system.....	28
Figure 4.3: Progress planning flow chart of the solar tracker project	29
Figure 4.4: 3D AutoCAD design of the framework	30
Figure 4.5: Front & side view AutoCAD design of the framework	31
Figure 4.6: Top view AutoCAD design of the framework.....	31
Figure 4.7: Final product of the solar tracker	32
Figure 4.8: AutoCAD design of the base supporting the motor	33
Figure 4.9: Final product of the driving motor	33
Figure 4.10: Block diagram scheme of the closed-loop tracker	34

Figure 4.11: Opaque objects besides the two sensors	34
Figure 4.12: Two sensors receive equal light intensity	35
Figure 4.13: One sensor receives more light intensity than the other	35
Figure 4.14: Comparison process between the two sensors	36
Figure 4.15: One LDR connected on top of the solar panel	36
Figure 4.16: Electric circuit diagram of the tracker system	37
Figure 5.1: Arduino platform with Integrated Development Environment.....	39
Figure 5.2: Analog readings of the light displayed through serial monitor.....	40
Figure 5.3: Tracking algorithm used in programming the microcontroller.....	41
Figure 6.1: Sun path diagram for Buraydah city on the test days	43
Figure 6.2: Solar energy and surface meteorology data for Buraydah city.....	44
Figure 6.3: Pictures of testing the solar tracker	44
Figure 6.4: Graph of voltage, current and power versus day's hours of tracker solar panel.....	46
Figure 6.5: Graph of voltage, current and power versus day's hours of stationary solar panel	47
Figure 6.6: Comparison of voltages produced by tracker and stationary solar panel	84
Figure 6.7: Comparison of currents produced by tracker and stationary solar panel	84
Figure 6.8: Comparison of output power by tracker and stationary solar panel	49

List of Tables

Table 1.1: Concise summary of the researches done so far in solar tracking field	13
Table 3.1: Parameters of the used solar cell	16
Table 1.2: Electric characteristics of the LDR	18
Table 3.3: Technical specification of Arduino Uno	19
Table 3.4: Specification of the driving DC motor	23
Table 6.1: Measurement values of the solar tracker panel	45
Table 6.2: Measurement values of the stationary solar panel.....	45
Table 6.3: Comparison of the output power of the tracker and stationary panel	49

Nomenclature

Symbol	Meaning
V_{oc}	Open circuit voltage
I_{sc}	Short circuit current
P_{Max}	Maximum power point
I_{ph}	Light generated current
I_d	Diode current
P_{out}	Output power
FF	Fill factor
A	Solar cell area
I	Light intensity.
η	Efficiency
M	Solar cells in parallel
N	Solar cells in series
V_T	Total voltage form panel
W _p	Nominal Peak Power
W _p	Output power tolerance
V _{mpp}	Maximum Power Voltage
I _{mpp}	Maximum Power Current
V _{oc}	Open-Circuit Voltage
I _{sc}	Short-Circuit Current
H c	Solar Cell Efficiency
H m	Module Efficiency
τ (N-m)	Torque
<i>rpm</i>	Revelation per minute
<i>ω (rad/sec)</i>	Angular velocity

Abstract

Currently, generating electricity by solar energy is inefficient and costly. The energy extracted from a solar photovoltaic depends on solar insolation. For the extraction of maximum energy from the sun, the plane of a solar collector should always be normal to the incident radiation. In this graduation project, we seek to improve the solar system efficiency by designing and implementing an automatic solar tracking systems which will keep the solar panel aligned with the sun in order to maximize solar power extraction. This system tracks the maximum intensity of light by adjusting the solar panel to be normal to the incident light. This sun tracker system uses two light dependent resistors as sensors element to find the brightest point in the sky. The data and signal processing of light sensors are performed by a microcontroller based system which controls a rotating DC Motor. Performance of this system over the important parameters like solar radiation received on the collector, maximum hourly electrical power, efficiency gain, short circuit current, and open circuit voltage has been evaluated and compared with those for fixed tilt angle solar collector.

CHAPTER1

INTRODUCTION

1.1 Overview

Our primary source of clean, abundant energy is the sun. The sun deposits 120,000 TW of radiation on the surface of the Earth, far exceeding human needs even in the most aggressive energy demand scenarios. Covering 0.16% of the land on Earth with 10% efficient solar conversion systems would provide 20 TW of power, nearly twice the world's consumption rate of fossil energy and the equivalent 20,000 1-GWe nuclear fission plants. These figures illustrate the impressive magnitude of the solar resource, providing an energy stream far more potent than present-day human technology can achieve [1].

While solar energy is abundant all over the planet, the solar energy in Saudi Arabia is recognized as from the highest available energy in the world since it is located in the earth-sun belt area that has high potential solar energy as shown in Fig 1.1. Therefore, the conversion of solar energy into electrical energy is one of the most promising and challenging technologies in K.S.A.

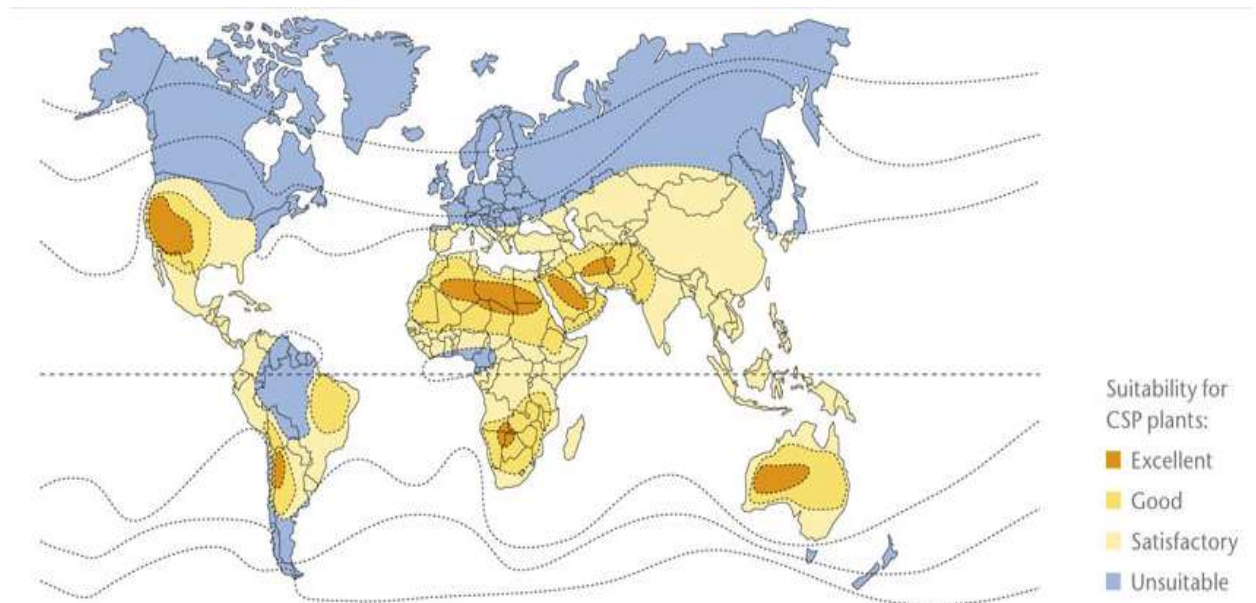


Fig.1.1: Earth-sun belt area with high potential solar energy [2]

Being clean, silent and reliable, with very low maintenance costs and minimal ecological impact are the main characteristics that made solar energy conversion favorable over other sources of energy. It is free, practically inexhaustible, and involves no polluting residues or greenhouse gases emissions. The technologies of harnessing solar energy are in continuous development and is utilized by a range of ever evolving technologies such as solar photovoltaics, solar heating, solar thermal electricity, solar architecture and artificial photosynthesis.

Due to the nature of solar energy, two components are required to have a functional solar energy generator. These two components are a collector and a storage unit. The collector simply collects the radiation that falls on it and converts a fraction of it to other forms of energy (either electricity and heat or heat alone). The storage unit is required because of the non-constant nature of solar energy; at certain times only a very small amount of radiation will be received. At night or during heavy cloud cover, for example, the amount of energy produced by the collector will be quite small. The storage unit can hold the excess energy produced during the periods of maximum productivity, and release it when the productivity drops. The technical feasibility and economic viability of these solar energy generator depends on the amount available sunlight (solar radiation) in the area where they intended to be placed.

The most popular technology used to convert solar energy to electrical energy is photovoltaics. It is a method of generating electrical power by converting sunlight into direct current electricity using semiconducting materials that exhibit the photovoltaic effect such as Si. The efficiency of these photovoltaic systems can be increased through three main ways. The first way is to increase the efficiency of a solar cell through manufacturing such as using efficient materials. The second is to maximize the energy conversion from the solar panel by developing a method whereby maximum power can be obtained from the voltage and current multiplied together. This is known as the maximum power point tracking (MPPT) and there are many developed algorithms that have been deployed to maximize the output power. The third way is by utilizing solar tracking system to maximize the incident power on the solar panel.

1.2 Problem Statement

Electrical energy from PV panels is derived by converting energy from the rays of the sun into electrical current. The main challenge is to maximize the capture of the rays of the sun upon the solar panels, which in turn maximize the output of electricity. A practical way of achieving this is by positioning the panels such that the rays of the sun fall perpendicularly on the solar panels by tracking the movement of the sun using an automatic solar tracker. Automatic solar tracker is an electro-mechanical system that increases power output of solar panel 30% to 60% than the stationary system. Therefore, a solar panel receives more sunlight when it is perpendicular or parallel to the sun, but the direction of the sunlight is always changes depends on the movement of the sun in a day. Mostly of the existed solar panels were installed statically, means that it does not moves to follow the position of the sun. In addition, most existed solar tracker are expensive and consist of many components that are hard to find especially in Saudi Arabia. Based on that, this solar tracker system project is to be designed and implemented to solve the previous problems.

1.3 Project objectives

The goal of this project is to design and implement a solar tracker system that can maximize the utilization of solar energy. The construction of this solar tracker system is to be inexpensive, requires low maintenance and overall needs to be more cost efficient. Also one of our objectives is to have an efficient tracking system so it take less energy than produced by the system. It is a closed-loop tracking system using sensor-based feedback controllers in which light dependent resistors (LDRs) are utilized to sense the sun's position and feedback error signal is generated to the control system to adjust the PV panel to an optimum position perpendicular to the solar radiation. It is a single-axis tracker with one degree of freedom follows the Sun's movement from the east to the west during a day.

The performance and characteristics of this automatic solar tracker are to be experimentally analyzed in Buraydah City. Buraydah city is located at the central region of Saudi Arabia (26.3333° N, 43.9667° E). It is characterized by the abundance of solar energy and one of the hottest regions in the world. The temperature in Buraydah could reach to 45⁰ in the summer. Figure 1.2 shows the temperature data or Buraydah taken from the world weather information agency and drawn in Excel software.

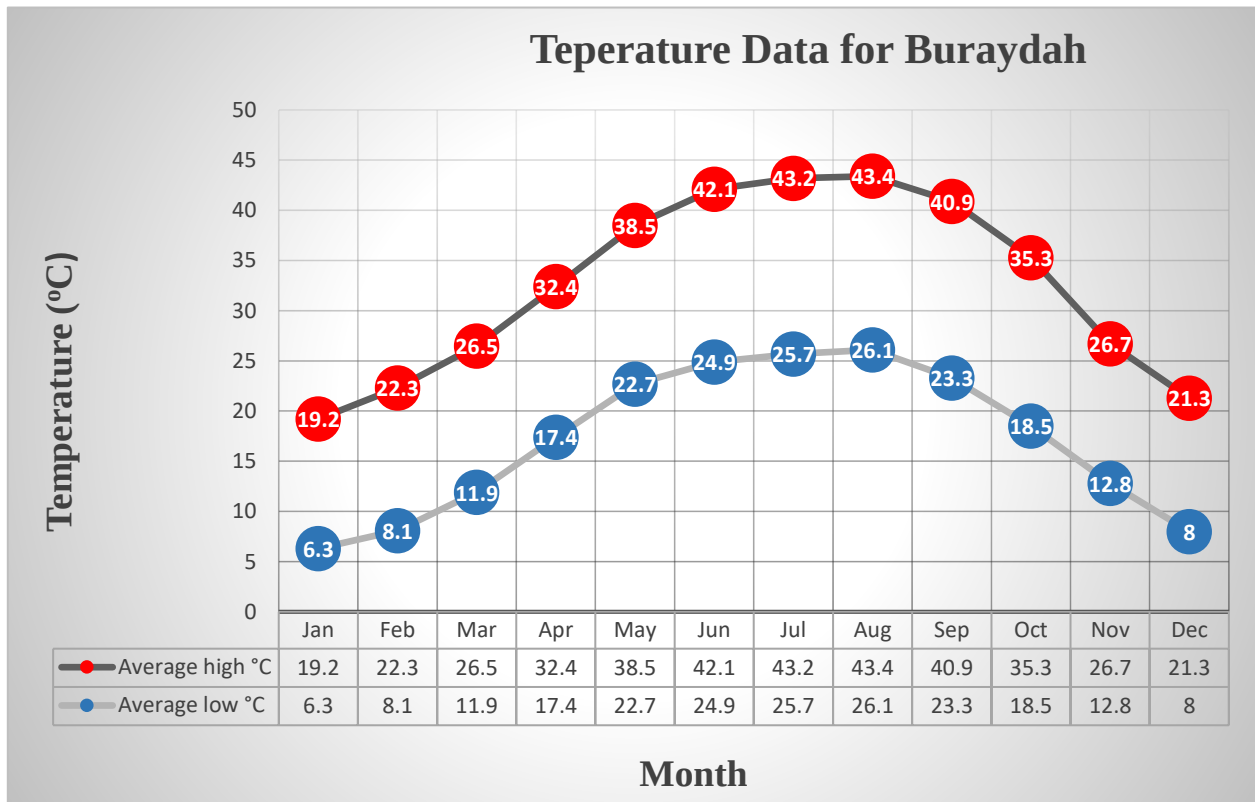


Fig. 1.2: Monthly temperature data for Buraydah City

1.4 Report Organization

This report for our graduation project consists of seven chapters. This chapter is an introductory overview of the project, objective of the project, problem statement, and report organization.

Chapter 2 presents a literature review of photovoltaics system and solar tracker systems. It contains a detailed description of solar photovoltaics grouping, working principles of solar cell, and a PV mathematical background. In addition, describes the previous research and projects done so far in the field of solar tracking systems. Overall, this chapter is about the basic knowledge on solar tracker.

Chapter 3 is about the hardware used in this project. It include a brief description on each components with their most considered electrical or mechanical characteristics.

Chapter 4 is discussing about the methodology that involves in the project design. It will explain how the project is organized and the flow of process in completing this project. It explains the methodology of project planning, assembly plan, circuit

Chapter 5 includes the software methodology. This will discuss more about the software that is used to program the microcontroller. Tracking Algorithms and the Arduino Code is presented as the two main components of software controlling.

Chapter 6 will discuss more about the result and discussion. This chapter will explains the process of testing our solar tracker by measuring the output power. The results then presented and a comparative analysis is done on the results of testing a stationary solar panel and the tracker panel.

This report is concluded with a brief conclusion that summarize what we have done and a dedicated section to the possible future improvements that can be done to improve the solar tracker further more.

CHAPTER2

Literature Review

2.1 Overview

Solar tracking system project had been widely employed by the other giant company like BP Solar, Yingli Green Energy, Kyocera, Q-Cells, Sanyo, Sharp Solar, Solar World, Sun Power, and Suntech. Now, many people use solar energy or photovoltaic energy as an alternative power because it's free and renewable. Photovoltaic energy is the most promising and popular form of solar energy. In solar photovoltaics', sunlight is actually converted into electricity. This is very different from a conventional understanding of solar power as only a way of heating water. Photovoltaic, now the biggest usage of solar energy around the world.

A photovoltaic system is a power system designed to supply usable solar power by means of photovoltaics. It consists of an arrangement of several components, including solar panels to absorb and directly convert sunlight into electricity, a solar inverter to change the electrical current from DC to AC, as well as mounting, cabling and other electrical accessories to set-up a working system. It may also use a solar tracking system to improve the system's overall performance or include an integrated battery solution. Photoelectric effect was first noted in 1839 by French physicist, Edmund Becquerel. He found that certain materials have property to produce small amounts of electric current when exposed to sunlight. In 1905, Albert Einstein described the nature of light and the photoelectric effect which has become the basic principle for photovoltaic technology. The first photovoltaic module was built by Bell Laboratories in 1954 [4].

2.2 Photovoltaic System

2.2.1 Photovoltaic cell

A solar cell is a solid-state electrical device (p-n junction) that converts the energy of light directly into electricity (DC) using the photovoltaic effect. When illuminated by sunlight it generates electric power. Solar cells are made up of various semiconductor materials such as: Mono-crystalline Silicon, Polycrystalline Silicon, Micro-crystalline Silicon, Cadmium Telluride (CdTe), and Copper Indium Selenide. But mono-crystalline silicon and poly-crystalline silicon are mainly used for commercial use. Figure 2.1 shows a diagram of the composition of a photovoltaic cell.

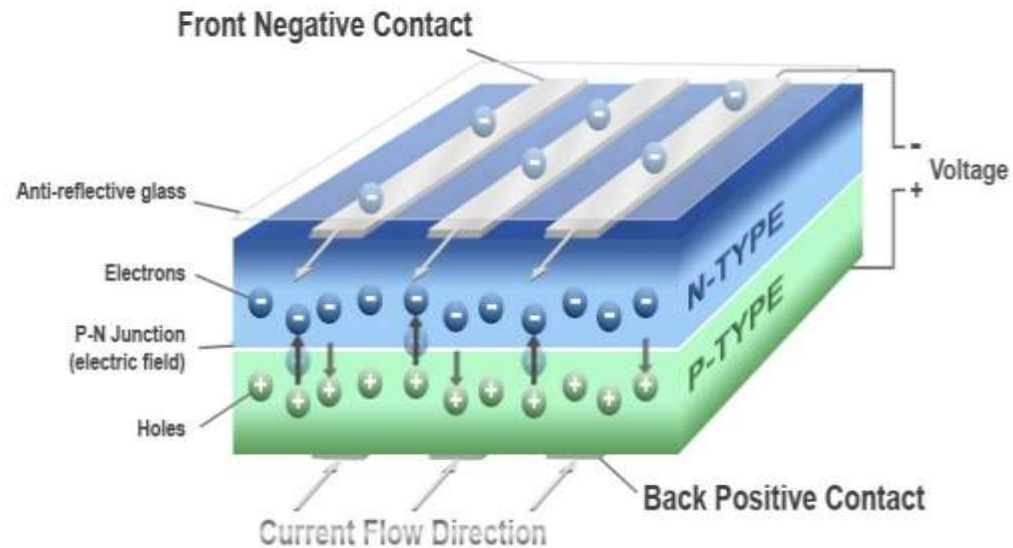


Fig. 2.1: Schematic diagram of the composition of a photovoltaic cell [5]

2.2.2 Photovoltaic module

The power produced by a single PV cell is not enough for general use. The solar cell is the basic unit of a PV system, but the power produced by a single solar cell is not enough for general use. An individual solar cell produces direct current and power typically between 1 and 2 W, hardly enough to power most applications. For example, in case of crystalline silicon solar cells with a typical area of $10 \times 10 \text{ cm}^2$, an output power is typically around 1.5 Wp, with $V_{oc} \approx 0.6 \text{ V}$ and $I_{sc} \approx 3.5 \text{ A}$ [4]. For actual usage, the solar cells are interconnected in series (for high voltage requirement) and in parallel (for high current requirement) combinations to form a PV module. Generally commercial modules consist of 36 or 72 cells. The modules consist of transparent front side, encapsulated PV cell and back side. The front side material is usually made up of low-iron and tempered glass. The efficiency of a PV module is less than a PV cell. This is due to the fact that some radiation is reflected by the glass cover and frame shadowing etc [6]. Figure 2.2 shows a diagram of a typical solar panel module with 36 cells connected in series.

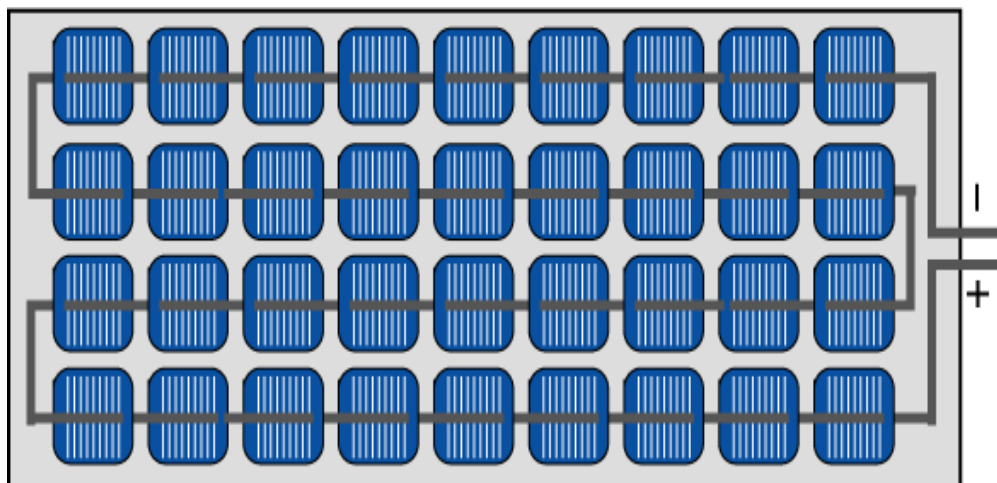


Fig. 2.2: Typical solar panel module with 36 cells connected in series [7]

2.2.3 Photovoltaic Array

A photovoltaic array (PV system) is an interconnection of modules which in turn is made up of many PV cells in series or parallel. The power produced by a single module is seldom enough for commercial use, so modules are connected to form array to supply the load. The connection of the modules in an array is same as that of cells in a module. Modules can also be connected in series to get an increased voltage or in parallel to get an increased current. Figure 2.3 shows a photovoltaic Array hierarchy.

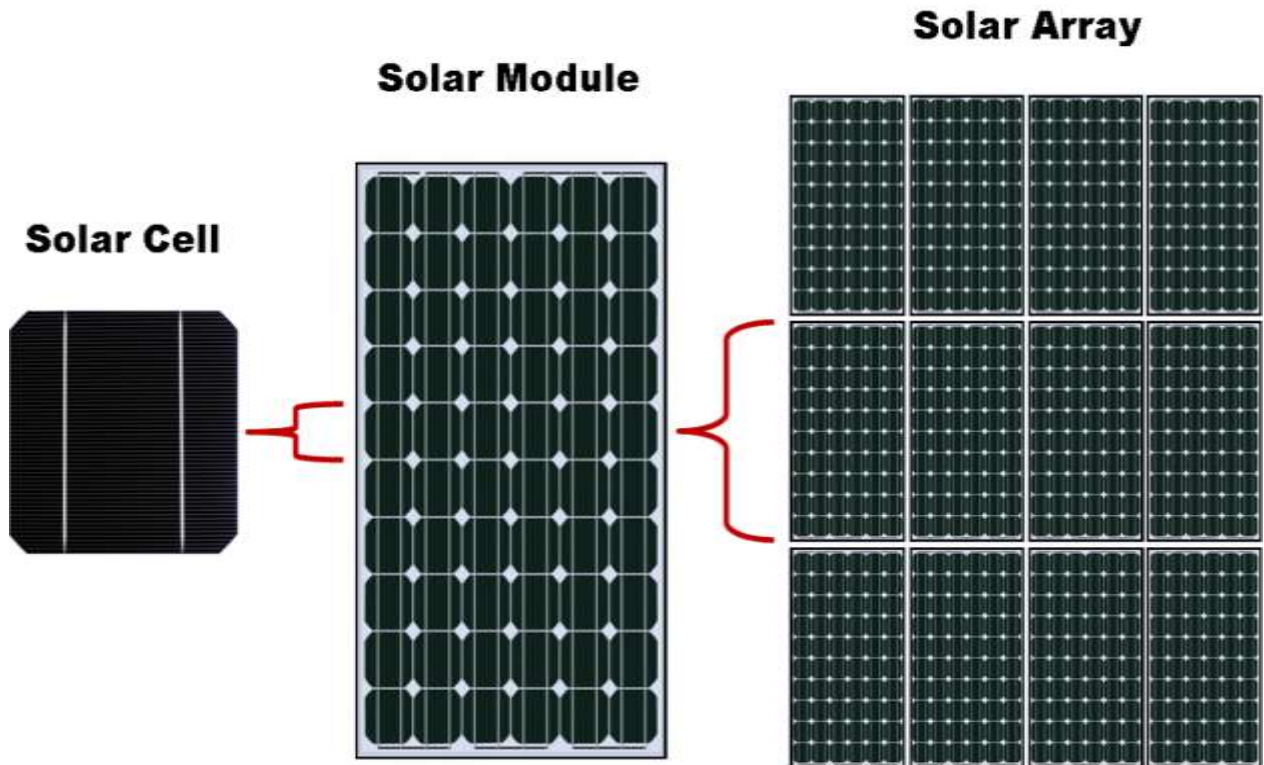


Fig. 2.3: Photovoltaic Array hierarchy [8]

2.3 Working of PV cell

Working of a PV cell is based on the basic principle of photoelectric effect. Photoelectric effect can be defined as a phenomenon in which an electron gets ejected from the conduction band as a consequence of the absorption of sunlight of a certain wavelength by the matter. So, in a photovoltaic cell, when sunlight strikes its surface, some portion of the solar energy is absorbed in the semiconductor material. If absorbed energy is greater than the band gap energy of the semiconductor, the electron from valence band jumps to the conduction band. By this, pairs of hole-electrons are created in the illuminated region of the semiconductor. The electrons thus created in the conduction band are now free to move. These free electrons are forced to move in a particular direction by the action of electric field present in the PV cells. These flowing electrons constitutes current and can be

drawn for external use by connecting a metal plate on top and bottom of PV cell. This current and the voltage (created because of its built-in electric fields) produces required power. Figure 3.3 shows the working principles diagram of solar panel. Figure 2.4 shows a diagram of the working principle of a PV cell.

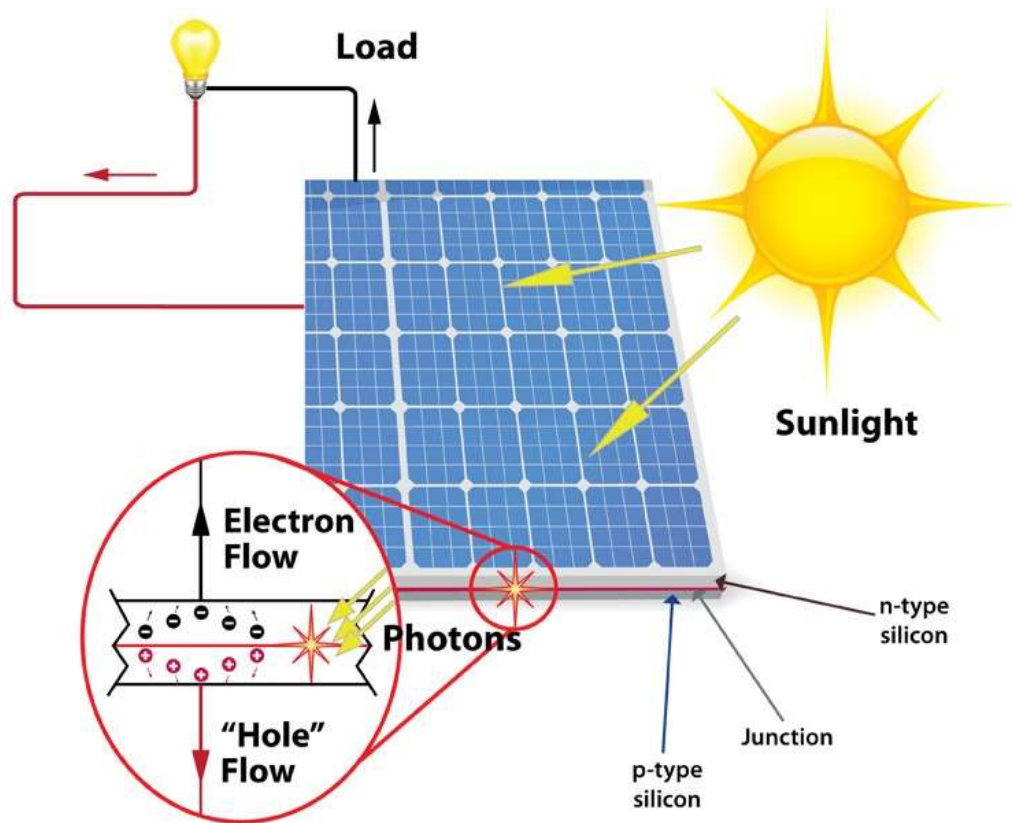


Fig. 2.4: Diagram of the working principle of a PV cell [9]

2.4 PV Mathematical Background

In solar cell operation, there are three important parameters to be considered: open circuit voltage V_{OC} , short circuit current I_{SC} and maximum power point P_{Max} . The maximum power that can be extracted from a PV cell are at the maximum power points. Usually manufacturers provide these parameters in their datasheets for a particular PV cell or module. The power calculation of a solar panel depends on the I-V characteristics of the solar cells it consists of. The I-V curve of a solar cell is the superposition of the I-V curve of the solar cell diode in the dark with the light-generated current.

The equation of the current in a solar cell in the dark is the same as the p-n junction:

$$I_d = I_0 \left[\exp \left(\frac{qV}{nkT} \right) - 1 \right]$$

The effect of the light on a solar cell is to introduce a light generated current I_{ph} opposite to the direction of the diode current I_d . Figure 2.5 shows a diagram of a solar cell diode in a short circuit and when there is a load [10].

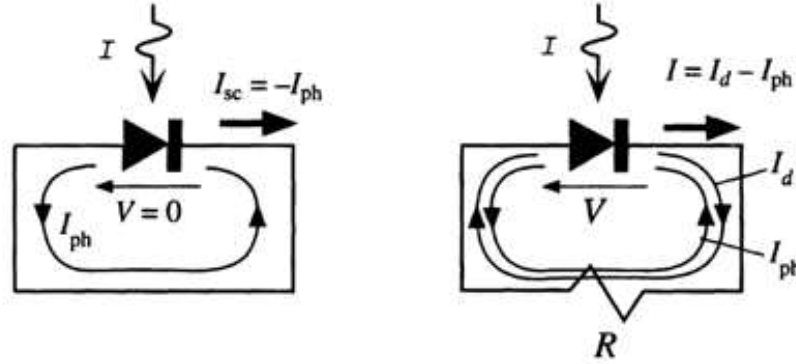


Fig. 2.5: Solar cell diagram in a short circuit and loaded circuit

The light generated current I_{ph} is dependent on the light intensity and on the properties of the solar cell. The general equation for calculating I_{ph} is:

$$I_{ph} = KI$$

Where: I is the light intensity, K is constant dependent on the particular solar cell type.

The total current of a solar cell is given by [10]:

$$I = I_d - I_{ph} = I_0 \left[\exp \left(\frac{qV}{nkT} \right) - 1 \right] - I_{ph}$$

The efficiency of a solar cell is the ratio of the output power to the incident power from the sun.

The output power is given by

$$P_{out} = V_{OC} I_{SC} FF$$

where: V_{OC} : Open-circuit voltage, I_{SC} : short-circuit current which is the same as the light generated current I_{ph} , FF : fill factor which depends on the solar cell and the loads connected.

The efficiency is then given by the equation:

$$\eta = \frac{P_{out}}{P_{in}} = \frac{V_{oc} I_{sc} FF}{P_{in}}$$

Where: P_{in} is the power from the sun incident on the solar cell, and it is calculated by:

$$P_{in} = A I$$

A : is the area of the solar cell, I is the light intensity.

Figure 2.6 shows I-V characteristic diagram of a solar cell.

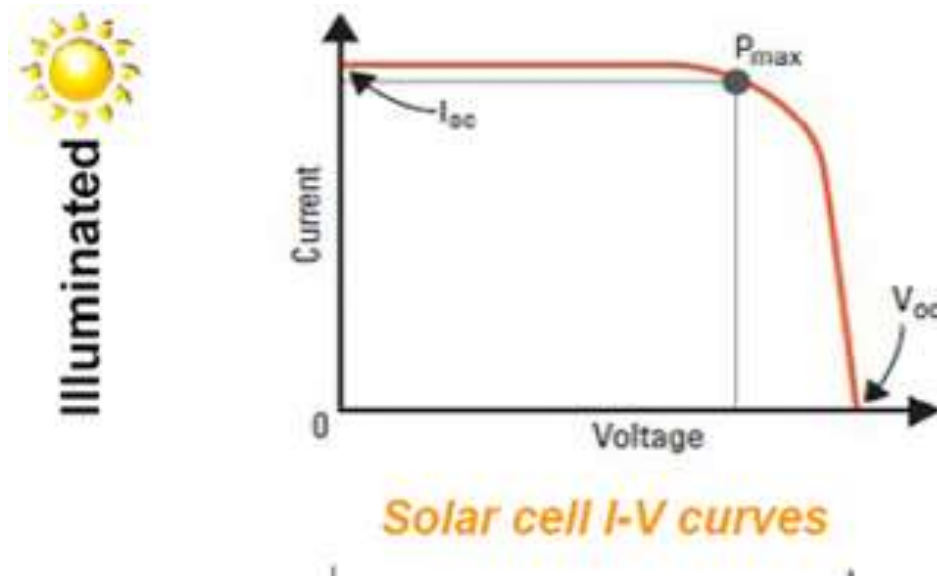


Fig. 2.6: I-V characteristic diagram of a solar cell [10]

The I-V characteristics of one module is the same as one solar cell, except that the voltage and current are increased. The total outage current of a solar panel is given by:

$$I = M I_d - M I_{ph} = M I_0 \left[\exp \left(\frac{q V_T}{N n k T} \right) - 1 \right] - M I_{ph}$$

Where: M is the number of cells in parallel, N is the number of cells in series; V_T is the total voltage from the panel.

2.5 Previous Solar Trackers

Many sun tracking systems have been proposed to facilitate the tracking of the sun's position with high degree of accuracy over the past 20 years. This section provides a literature overview of the sun tracking system field with description of the more significant proposal for and open loop types of sun tracking systems. Table 1.1 shows a concise summary of the researches done so far in solar tracking field.

Table 1.1: Concise summary of the researches done so far in solar tracking field

Scholar's Name	Description of The Sun-Tracking System	Date
Maish	Developed a control system called SolarTrak to provide sun tracking, night and emergency storage, communication, and manual drive control functions for one- and two-axis solar trackers in a low-cost, user-friendly package. The experimental results showed that the control system enabled a full-day pointing accuracy of better than $+0.1^\circ$ to be achieved [11].	1990
Agarwal	Presented a two-axis tracking system consisting of worm gear drives and four bar-type kinematic linkages to facilitate the accurate focusing of the reflectors in a solar concentrator system [12].	1992
Brown and Stone	Developed a tracking system for solar concentrators in which a neural network was applied to an error model in order to compensate for tracking errors. The test data showed that the resulting system was capable of reducing the tracking error to a value of less than 0.01° [13].	1993
Kalogirou	Presented a one-axis sun-tracking system utilizing three light-dependent resistors (LDRs). The output signals from the three LDRs were fed to an electronic control system which actuated a low-speed 12 - V DC motor in such a way as to rotate the collector such that it remained pointed toward the sun [14].	1996
Khalifa and Al-Mutawalli	Developed a two-axis sun tracking system to enhance the thermal performance of a compound parabolic concentrator. It was shown that the tracking system had a power consumption of just 0.5 Whr and yielded an improvement of around 75% in the collected solar energy, compared to a fixed collector of equivalent dimensions[15].	1998
Yousef	Developed a sun tracking system in which the nonlinear dynamics of the tracking mechanism were controlled using a fuzzy logic control algorithm implemented on a PC and supported by an interfacing card consisting of a sensor data acquisition function, motor driving circuits, signal conditioning circuits and serial communications [16].	1999

Falbel et al.	Presented a sun-oriented attitude-control system combined with a concentrating solar panel for use in a satellite. The calibration results showed that the sensor was capable of locating the position of the sun with an accuracy of $+0.05^\circ$ [17].	2002
Urbano et al.	Presented a 5 Watt-PV module for a stand-alone solar tracking system with a capacity of 2.6 kW. The tracking system was designed to follow the position of the sun autonomously in the altitude and azimuth directions and was driven by two 12 V DC motors, each with a power consumption of 36 W and both fed by a single electrolytic condenser charged by the PV module [18].	2003
Jiang and Cao	Constructed an emulated sunflower based on a spherical four-quadrant photoelectric sensor for solar tracking purposes [19].	2003
Luque-Heredia et al.	Presented a sub-degree precision sun tracker for 1,000X micro-concentrator modules. The tracking system comprised a lightweight structure designed to remain operative for high wind speeds, yielded 95% of the available direct solar radiation and featured an electronic tracking control unit which relied on an adaptive algorithm for absorbing unforeseen or time varying errors with automatic calibration of cheap sun pointing sensors against array power output [20].	2003
Berenguel et al.	Developed an automatic heliostat offset correction control system based on an artificial vision technique and common charge-coupled device (CCD) equipment [21].	2004
Al-Mohamad	Used a programmable logic controller (PLC) to control a photovoltaic module for following the sun's radiation. It was shown that by collecting and storing the data relating to the sun's radiation, and using this information to control the photovoltaic module, the daily output power of the photovoltaic system could be improved by more than 20% relative to that obtained from a fixed module [22].	2004
Alata et al.	Designed and simulated three sun tracking systems, namely: (1) one-axis sun tracking with the tilted aperture equal to the latitude angle, (2) equatorial two-axis sun tracking, and (3) azimuth/ elevation sun tracking. For each tracking system, the modeling and controller design tasks were accomplished using the first-order Sugeno fuzzy inference system [23].	2005
Chen et al.	Presented a sun sensor algorithm based on an analogue optical nonlinear compensation measuring principle. The experimental results showed that the proposed sensor had an accuracy of better than 0.2° over the entire field of view of $+62^\circ$ for both axes [24].	2007

CHAPTER 3

PROJECT HARDWARE

3.1 Monocrystalline solar panels

In this project we used monocrystalline solar panels (model: 5432/100W), which is relatively inexpensive panel. These panels are manufactured by using crystalline silicon, a basic semiconductor material that is produced in large sheets which can be cut to a specific size. These panels are more expensive than the polycrystalline panels that follow. However, they are highly efficient and are more cost-effective in the long run as a result. Monocrystalline panels are typically 15-18% efficient. Figure 3.1 shows a picture of the solar panel used in this project.



Fig. 3.1: The solar cell panel used in this project

Table 3.1 shows the parameters of the used cell, and Figure 3.2 shows a picture of the parameters.

Table 3.1: Parameters of the used solar cell

Nominal Peak Power	W_p	100 W
Output power tolerance	W_p	+/-5%
Maximum Power Voltage	V_{mpp}	17.8 V
Maximum Power Current	I_{mpp}	5.62 A
Open-Circuit Voltage	V_{oc}	21.1 V
Short-Circuit Current	I_{sc}	6.07 A
Solar Cell Efficiency	H_c	13 %
Module Efficiency	H_m	12.5 %
Length x Width x Height		1210x660x35mm



Fig. 3.2: Cell parameters

3.2 Light Dependent Resistors

Light Dependent Resistor (LDR) is a resistor whose resistance is inversely dependent on the amount of light falling on its surface. LDR is made of Cadmium Sulfide (CdS) which is an active semiconductor layer that is deposited on an insulating substrate. The semiconductor layer is lightly doped to enable to have the required level of conductivity. If light falling on the device is of high enough frequency, photons absorbed by the semiconductor give bounded electron enough energy to jump into the conduction band. The resulting free electron (and its hole partner) conduct electricity, thereby lowering resistance. In our project, we used two (12mm CdS) LDRs as sensors for the light intensity as shown in Fig.3.3.

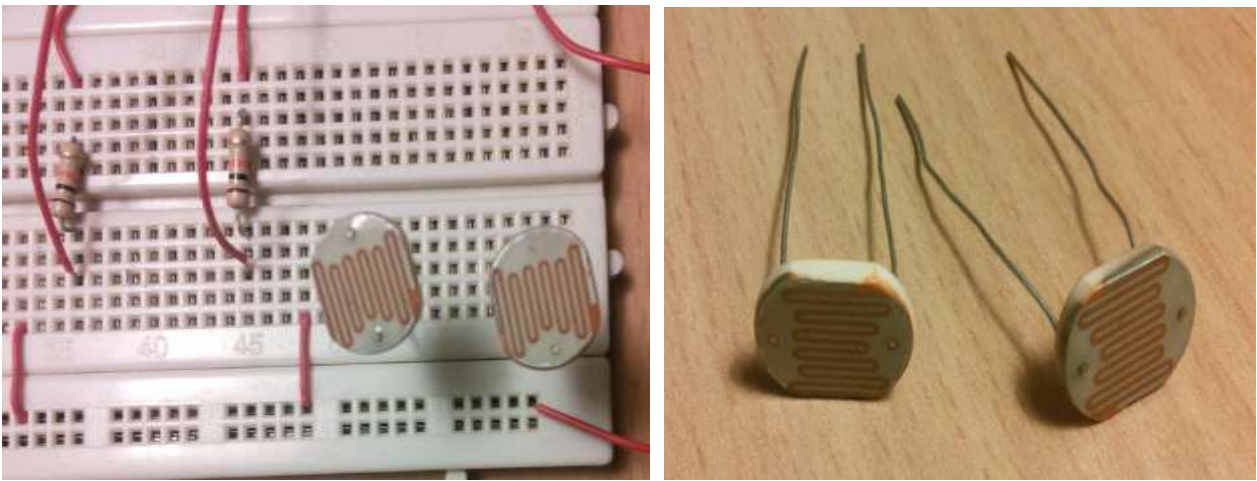


Fig. 3.3: (12mm CdS) light sensors

These LDRs are characterized by their quick response, reliable performance, Epoxy package, and good characteristics of spectrum. Figure 3.4 shows dimensions and components of one LDR.

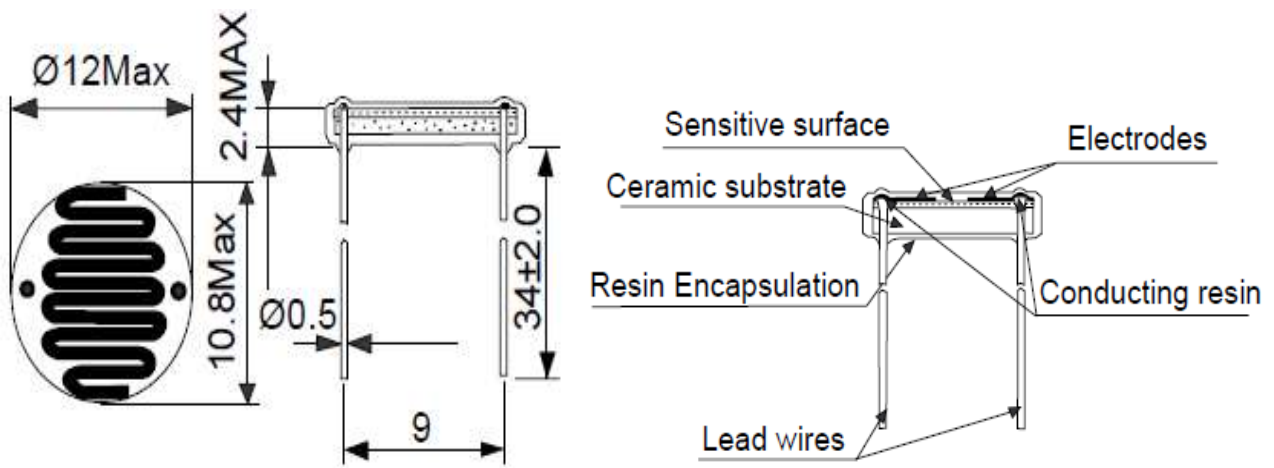


Fig. 3.4: Dimensions and components of one LDR [25]

In order to use these LDRs properly, the electric characteristics must be considered first, these characteristics are shown in Table 3.2. Figure 3.5 shows graph of the sensitivity of LDR to the wavelength of the incident light and a graph of the effect of temperature on LDR's resistance.

Table 3.2: Electric characteristics of the LDR

$V_{\max}$ (VDC)	$P_{\max}$ (mW)	Ambient Temp (°C)	Spectral Peak (nm)	Dark Resistance (M Ω)	Response Time (ms)	
					Rise	Decay
250	250	-30~+70	560	5.0	30	30

Where:

- $P_{\max}$: Maximum power dissipation at ambient temperature of 25 °C
- $V_{\max}$: Maximum voltage in darkness that may be applied to the device continuously.
- Spectral Peak: Spectral sensitivity of LDR depends on the wavelength of light they are exposed to and in accordance with figure “Spectral Response”.

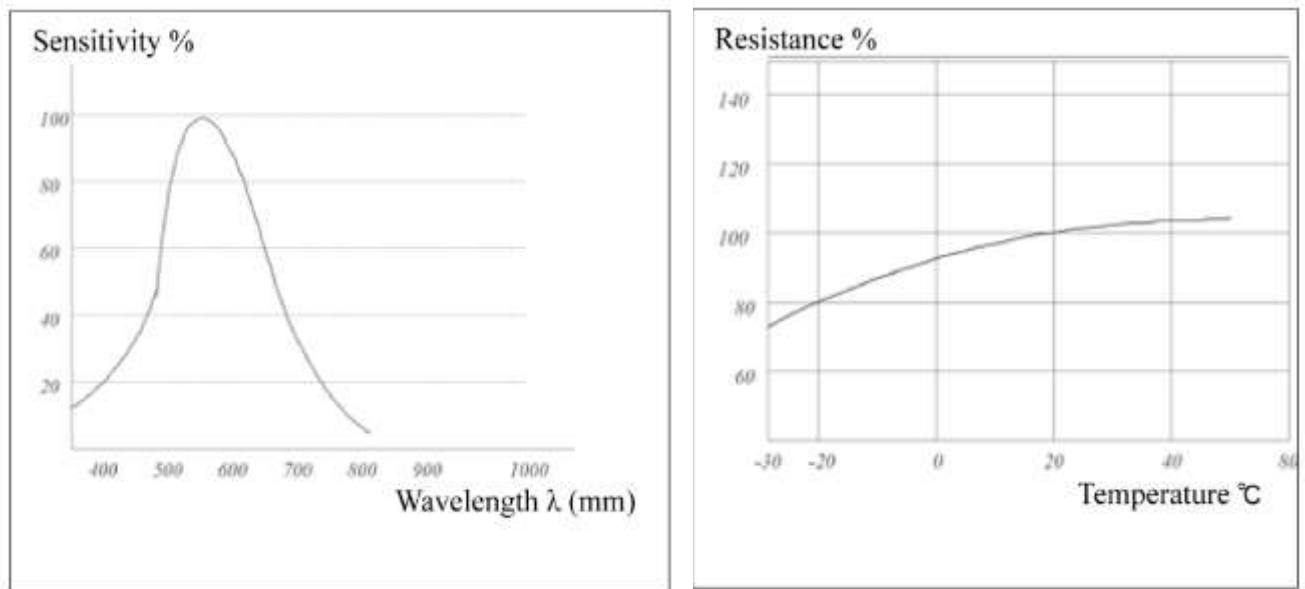


Fig. 3.5: Sensitivity of LDR to the wavelength of the incident light and the effect of temperature on LDR's resistance [25]

3.3 Microcontroller

To control the DC motor in automatic solar tracker, we used an Arduino Uno as a microcontroller to receive the signal from the tow photo sensor, and then control the operation of the DC Motor according to the light difference between the two sensors. Figure 3.6 shows the used Arduino Uno and its external power supply regulator.

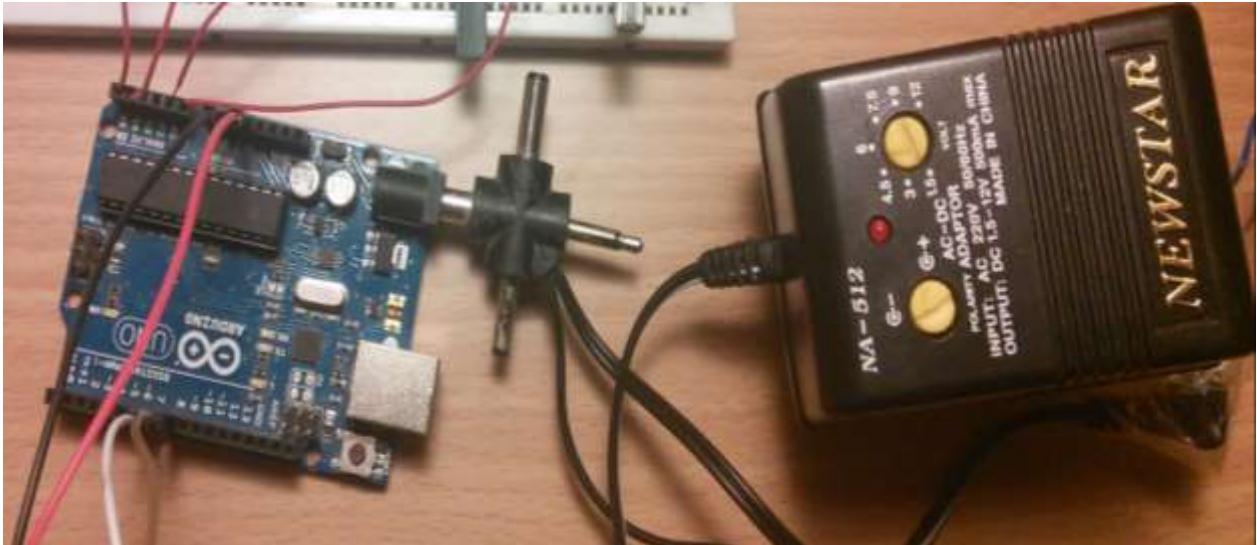


Fig. 3.6: Arduino Uno and its external power regulator.

Table 3.3 shows a summary of the technical specification of the used microcontroller (Arduino Uno)

Table 3.3: Technical specification of Arduino Uno [26]

Microcontroller	ATmega328
Operating Voltage	5V
Input Voltage (recommended)	7-12V
Input Voltage (limits)	6-20V
Digital I/O Pins	14 (of which 6 provide PWM output)
Analog Input Pins	6
DC Current per I/O Pin	40 mA
DC Current for 3.3V Pin	50 mA
Flash Memory	32 KB of which 0.5 KB used by bootloader
SRAM	2 KB
EEPROM	1 KB
Clock Speed	16 MHz

The Arduino Uno is an open-source microcontroller platform with development environment that implements Processing/Wiring language which is a subset of C and includes several libraries. It is a microcontroller board based on the ATmega328. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz crystal oscillator, a USB connection, a power jack, an ICSP header, and a reset button. It contains everything needed to support the microcontroller and it can be connected to a computer with a USB cable. Integrated USB connection can be used for simple FLASH ROM programming or for communication with other devices. Included library allows serial communication on any of the Uno's digital pins. The ATmega328 also supports I2C (TWI) and SPI communication. The Arduino software includes a Wire library to simplify use of the I2C bus. Figure 3.7 shows an Arduino Uno microcontroller with its labeled components.

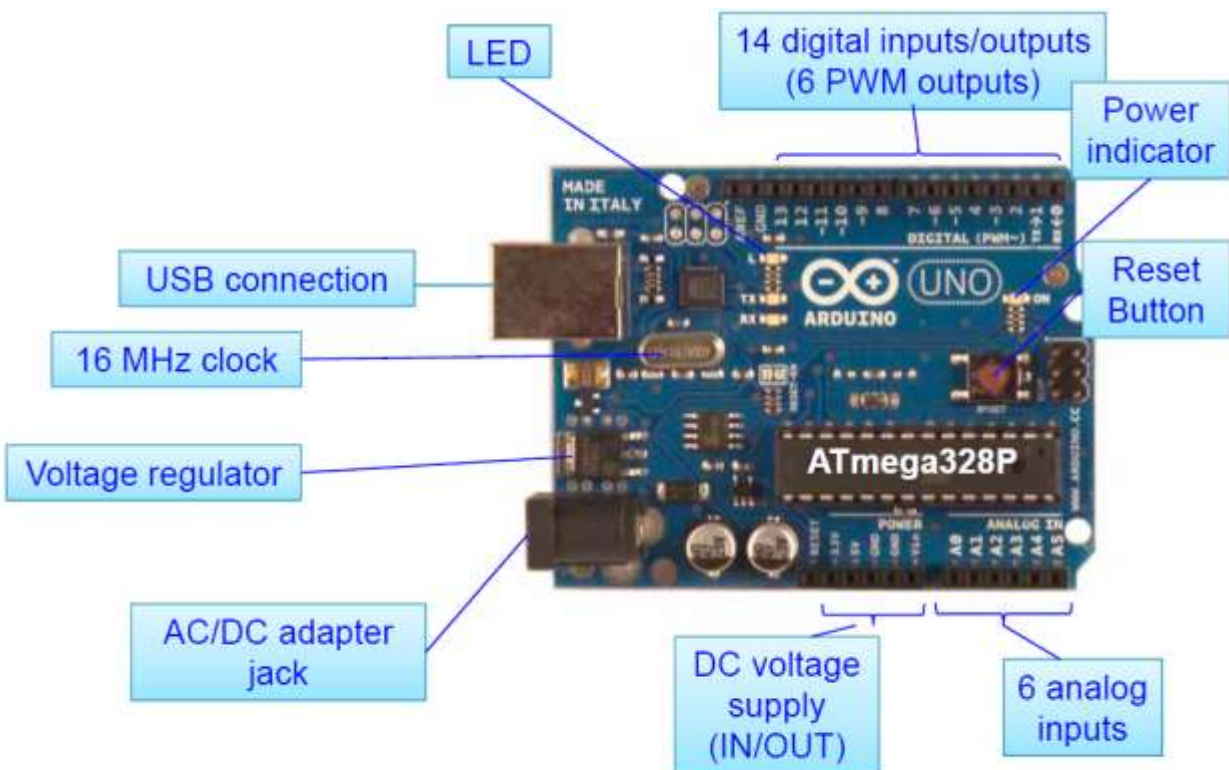


Fig. 3.7: Arduino Uno components

The Arduino Uno can be powered via the USB connection or with an external power supply. The power source is selected automatically. External (non-USB) power can come either from an AC-to-DC adapter (wall-wart) or battery. The adapter can be connected by plugging a 2.1mm center-positive plug into the board's power jack. Leads from a battery can be inserted in the Gnd and Vin pin headers of the POWER connector.

The Atmega328 has 32 KB of flash memory for storing code (of which 0,5 KB is used for the boot loader); It has also 2 KB of SRAM and 1 KB of EEPROM.

3.4 Relay Module

The microcontroller in our project is connected to 2 Relay module, which is connected to the battery from one side and the DC motor from other side. A relay is an electrical switch that uses an electromagnet to move the switch from the off to on position instead of a person moving the switch. It takes a relatively small amount of power from the Arduino Uno to control switching on and off of the DC motor that draws much power from the battery. Figure 3.8 shows a picture of used relay module.



Fig. 3.8: Relay module to be connected to a DC motor

Figure 3.9 shows the wiring and board setup of the relay module.

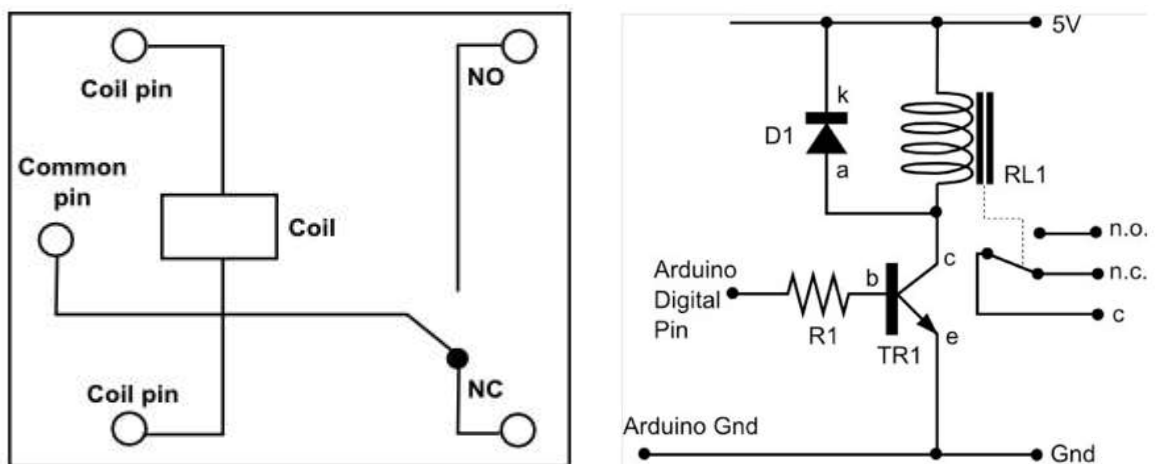


Fig. 3.9: Wiring and board setup of the relay module [27]

3.5 DC Motor

The motor used in this tracking system to drive the solar panel is used Windshield Wiper motor. The reason for choosing this motor is it inexpensive and gives sufficient torque to turn the panel. Figure 3.10 shows pictures of the used DC motor, while Fig. 3.11 shows its drawings with dimensions.



Fig. 3.10: Used windshield wiper DC motor driving solar panel

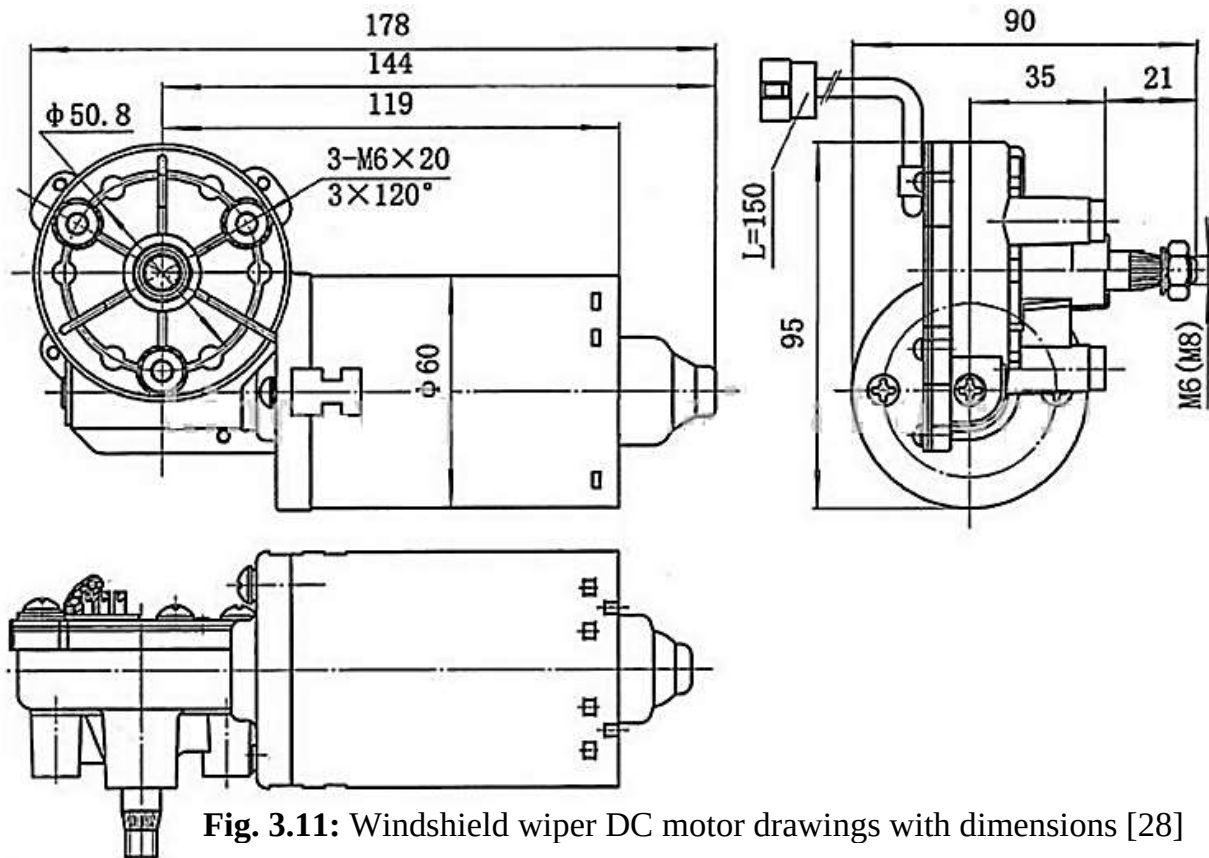


Fig. 3.11: Windshield wiper DC motor drawings with dimensions [28]

The specification of the motor is given in Table 3.4.

Table 3.4: Specification of the driving DC motor

Name	Ford Windshield Wiper Motor
Direction	Clockwise & anticlockwise (opposite polarities)
No Load High Speed	12 volts, 47 rpm, Stall Torque (hi): 3.02 N-m
No Load Low Speed	12 volts, Speed 38 rpm, Stall Torque (low) 5.03 N-m

The Current rating of this motor is calculated as follows:

Since the Motor is pulsed at a certain time interval so we assume at low speed, hence:

Power = Torque (T in N-m) $\times$ Speed (ω in rad/sec) = $5.03 \times (2\pi \times 38/60) = 20.0$ Watts.

Power = Voltage $\times$ Current = 20.0 Watts.

Current rating = $20.0/12 = 1.67$ Ampere.

3.6 Battery

We used a regular car battery to power the DC motor. This battery is to be charged by the solar cell using a charge controller. Figure 3.12 shows a picture of the battery powering the DC motor.

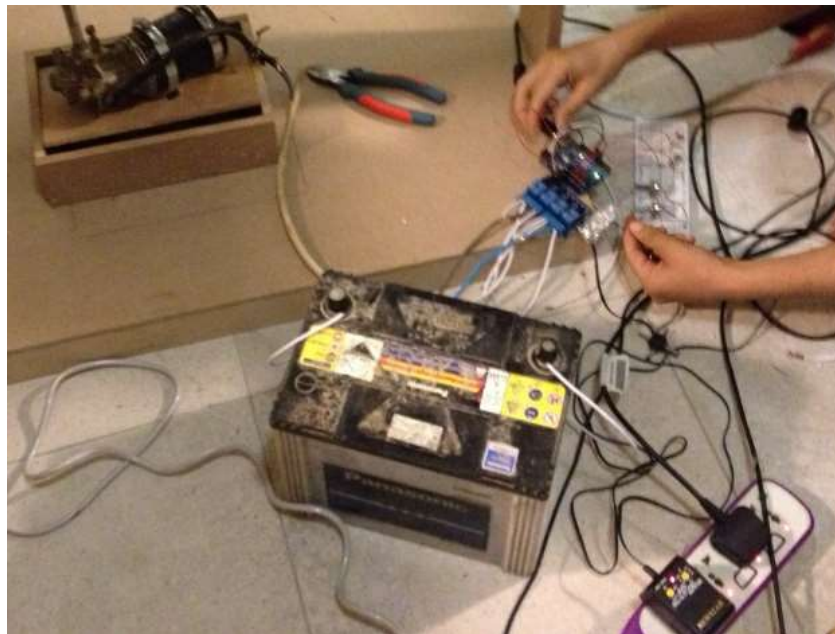


Fig. 3.12: Battery powering the DC motor

CHAPTER 4

METHODOLOGY

4.1 Overview

This chapter describes the methodology used and flow of steps used in the design and implementation of this tracking system. The project was divided into parts to make the design process modular. The project consists of reading a series of light sensor values, comparing them, and then positioning a motor to align with the greatest value which corresponds to the sun's position. In the design, we used the engineering design process throughout all phases of our project as shown in Fig. 4.1.

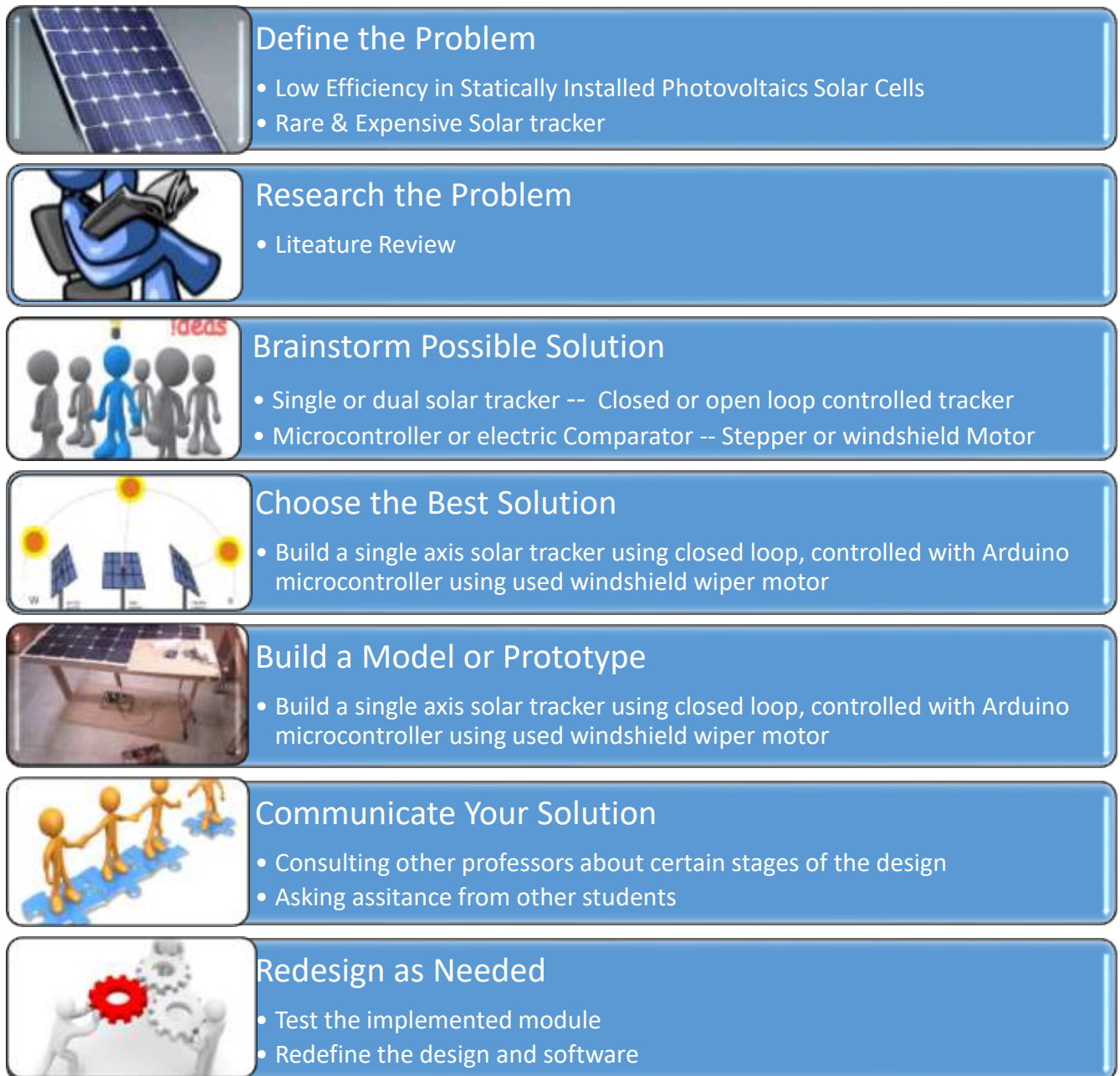


Fig. 4.1: Engineering design process used in the design and implementation of this tracker

4.2 Project Planning

In the process of project planning, first the problem is identified which is: low power efficiency in statically installed solar panels and expensive and rare existed solar tracker. Then the objective of the project is identified which is to design and implement inexpensive solar tracker. However, before deciding the course of action, we followed a strategic design process as explained in the following points:

Brain storming

The team gathered and brainstormed different ideas, some of these ideas are:

- Build a single axis solar tracker (horizontal, vertical, polar).
- Build a dual axis solar tracker (Tip-tilt, Azimuth-altitude).
- Build a single axis solar tracker that can be modified seasonally by using manual adjustment.
- Build a closed loop controlled solar tracker.
- Build an open loop controlled solar tracker.
- Use a microcontroller as the controlling equipment.
- Use an electric circuit as a comparator to control the solar tracker.
- Use stepper motor as the driving motor.
- Use used windshield wiper motor.

Deciding the course of action

In evaluating the possible solutions, we take into consideration that the proposed solution should satisfy these technical requirements.

- Minimum energy consumption, for the maximization of global efficiency of the installation and optimum performance-cost ratio.
- Reliability in operation, under different perturbation conditions (wind, dust, rain, important temperature variations).
- Simplicity of movement solution (motor, gears, sensors), to diminish the cost and to increase the viability.
- Possibility of system integration in a monitoring and control centralized structure, which means a digital control solution.

After evaluating every possible solutions, we decided to build a single axis tracker that can be modified seasonally by using manual adjustment since the sun's position change every season from north to south and vice versa. Our reasons are that it has the combination of energy improvement, low cost parts and low installation complexity. Also, we decided to use closed loop tracker system since it will produce a better tracking efficiency than open loop tracker as well as its performance are precise and observe the output of the process

that it is controlling. Windshield wiper motor is chosen to the driving motor, since it is inexpensive and can be found in any used car shop. Arduino Uno is, also, chosen to be the microcontroller since it can be managed easily and relatively inexpensive. In addition, we decided to use a used car battery due to its availability.

Constraints & Assumptions

In the beginning of the design process, we faced certain constraints and made some assumptions. Some of constraints are:

- The weight of the tracking system must be in a reasonable range, since the light weight will reduce the cost of the material as well as the power needed for tracking system itself.
- The new solar tracker needs to be built with small budget in mind, because there are some commercial solar tracking systems on the market but the costs are usually high.
- The built solar tracker cannot be fixed in one place, since this a project that needs to be moved around.

Besides these constraints, we made the following assumptions in the beginning of the project.

- The solar cell will operate in ideal weather conditions. That is we assume that there will be no sand storm, heavy rains, and extreme temperature conditions. These extreme weather conditions can destroy the solar cell, the supporting structure or the electric circuit.
- The electric circuit is powered from the solar system itself. We used an electric converter to convert from 12 volt to 5 volt to power the Arduino in which it power the electric circuit.
- The DC motor does not to be quick in responding to a controlling signal, since the light of the sun change its direction slowly.

Design Heuristics

After evaluating the different ideas and choosing the course of action, we proceeded to the design of the whole project. To design effectively, we visualized the general working principle of this solar tracker. Figure 4.2 shows the solar tracker project flow chart which served as a design heuristic tool that aided in the general design. The basic principle of the project is that, First, the sun emit the light and the sensors will detect the intensity of light and send the data the Arduino Uno microcontroller. Next, the microcontroller will compare the intensity of light based on the data that had been collected from sensors and send a signal to the driving motor. The driving motor will rotate the solar panel to the right direction based on the data that send by the microcontroller. The solar panel will charge the battery, so the system is self- powered system.

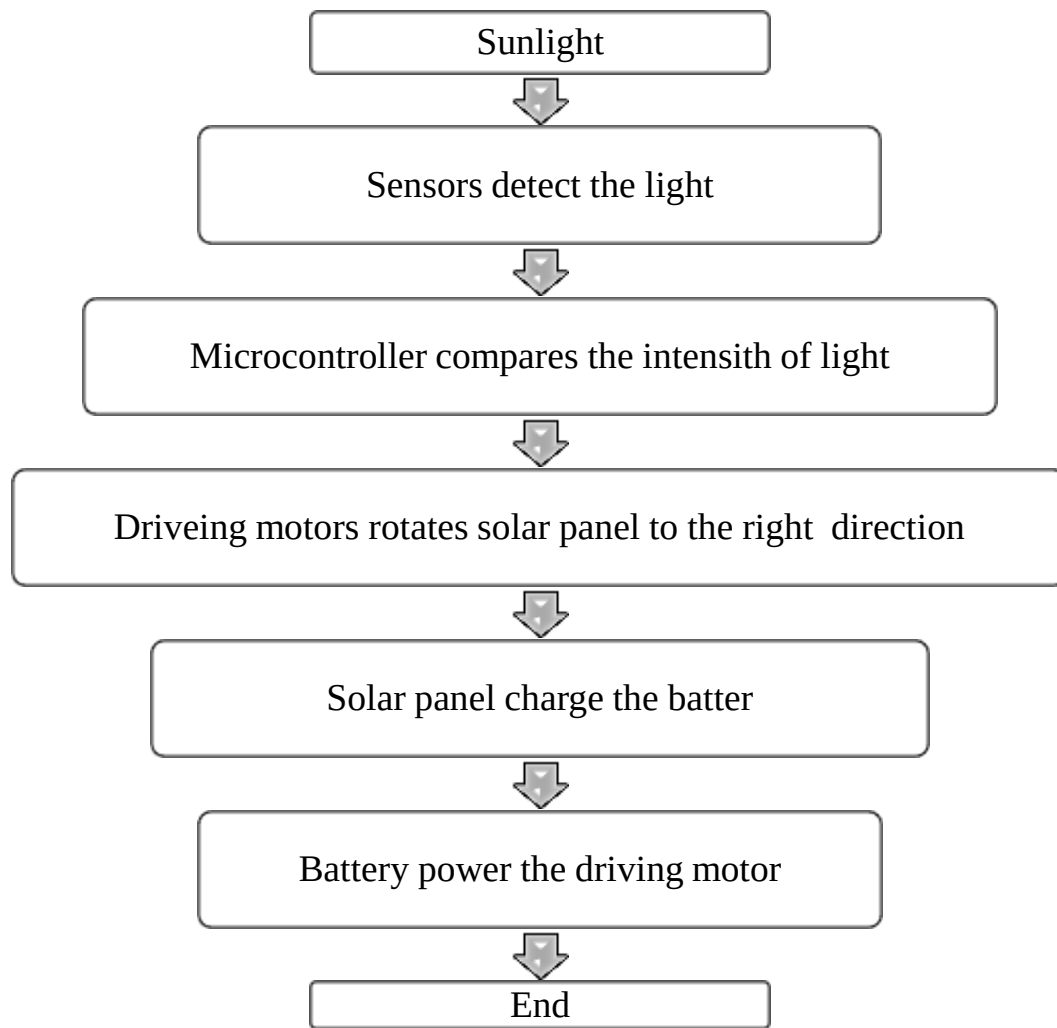


Fig. 4.2: Project flow chart of the solar tracking system

Design Process/Progress Planning

We progressed in our project by making a plan of the design process, this design process plan is summarized in flow chart shown if Fig. 4.3. First, the hardware must be purchase, tested to make sure they operate perfectly. The solar test is tested to ensure that it produce the specified output power, the DC motor is tested by using a car battery. The Arduino Uno, relay module, and LDRs are test separately, to ensure they work properly and prepare them for the actual use. Then the structure is to be designed and built with minimum cost. The final steps are the hardware implementation, software implementation and evaluating the final product. The solar cell is to be mounted on the solar and the electric circuit to be connected together. Finally, the microcontroller is to be programmed based on the system that we need to construct.

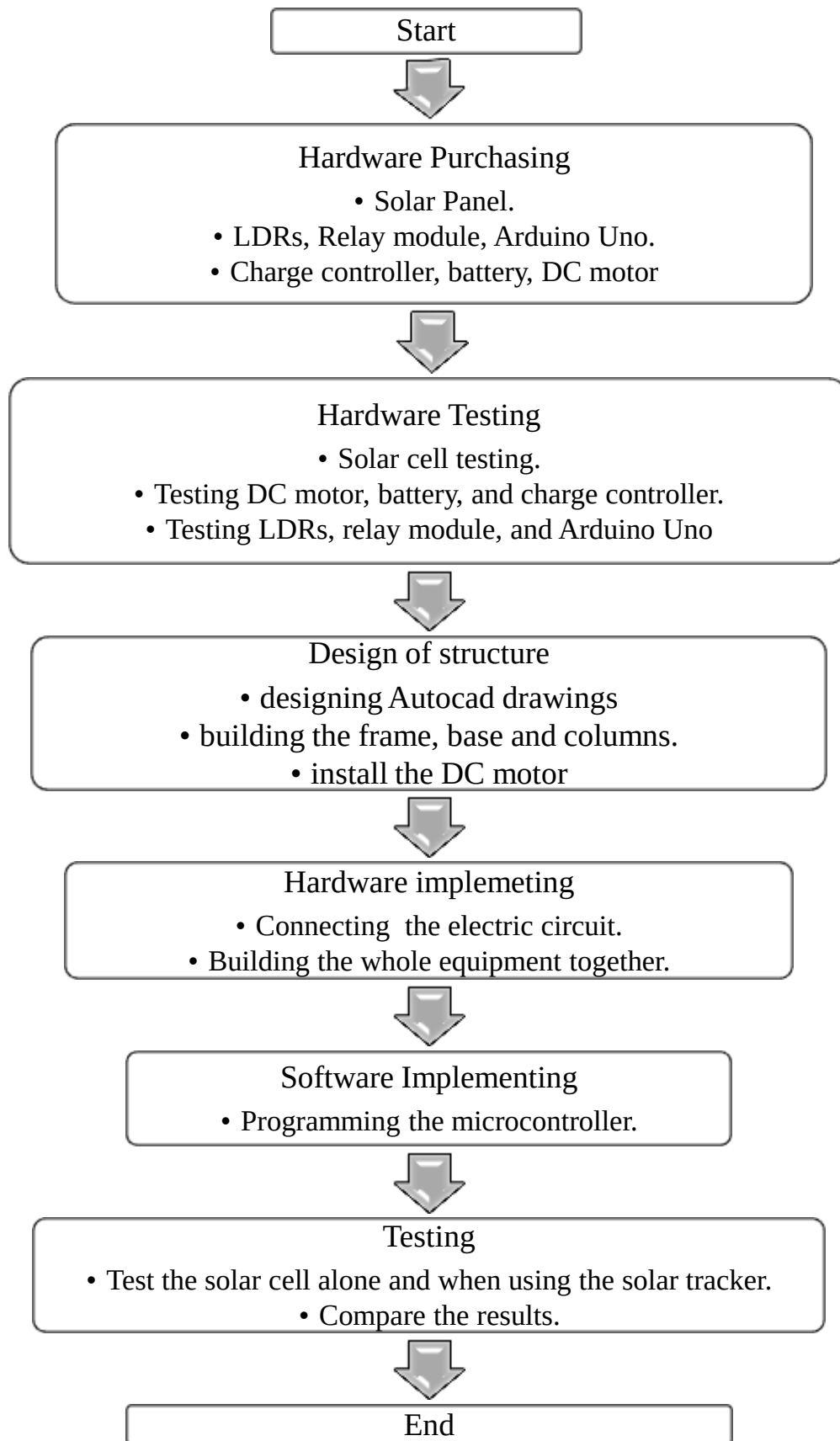


Fig. 4.3: Progress planning flow chart of the solar tracker project

2.3 Design of Framework

Solar tracker needs a supporting structure that can hold the solar panel and in the same time it can rotates. The best way was to build from a wood a base and two columns. The frame in which the panel is put at the top supported by the two columns through two screws that can rotates. This idea then designed using AutoCAD software before it was built to ensure that the dimensions and the specific details are made properly. Figure 4.4 shows the design in 3D (45 degree).

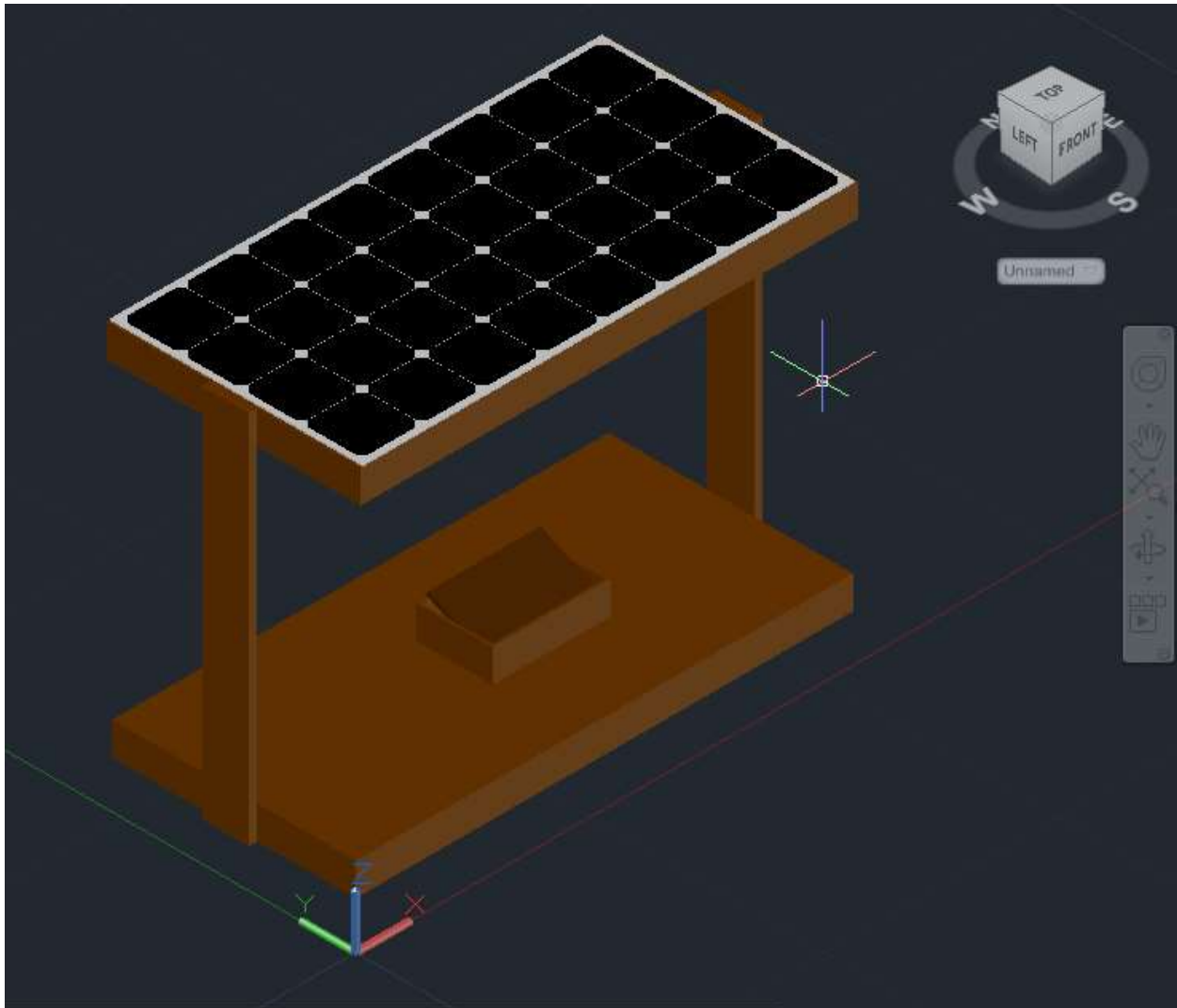


Fig. 4.4: 3D AutoCAD design of the framework

Figure 4.5-4.6 show the AutoCAD designs of the framework (front view, side view, and top view respectively).

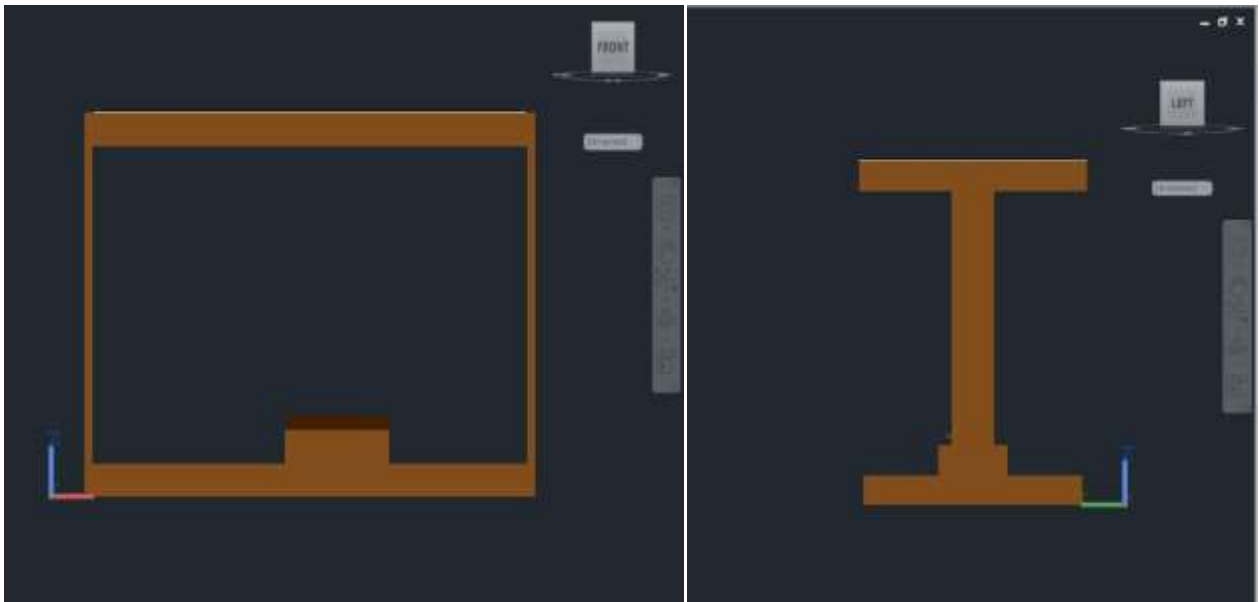


Fig. 4.5: Front & side view AutoCAD design of the framework

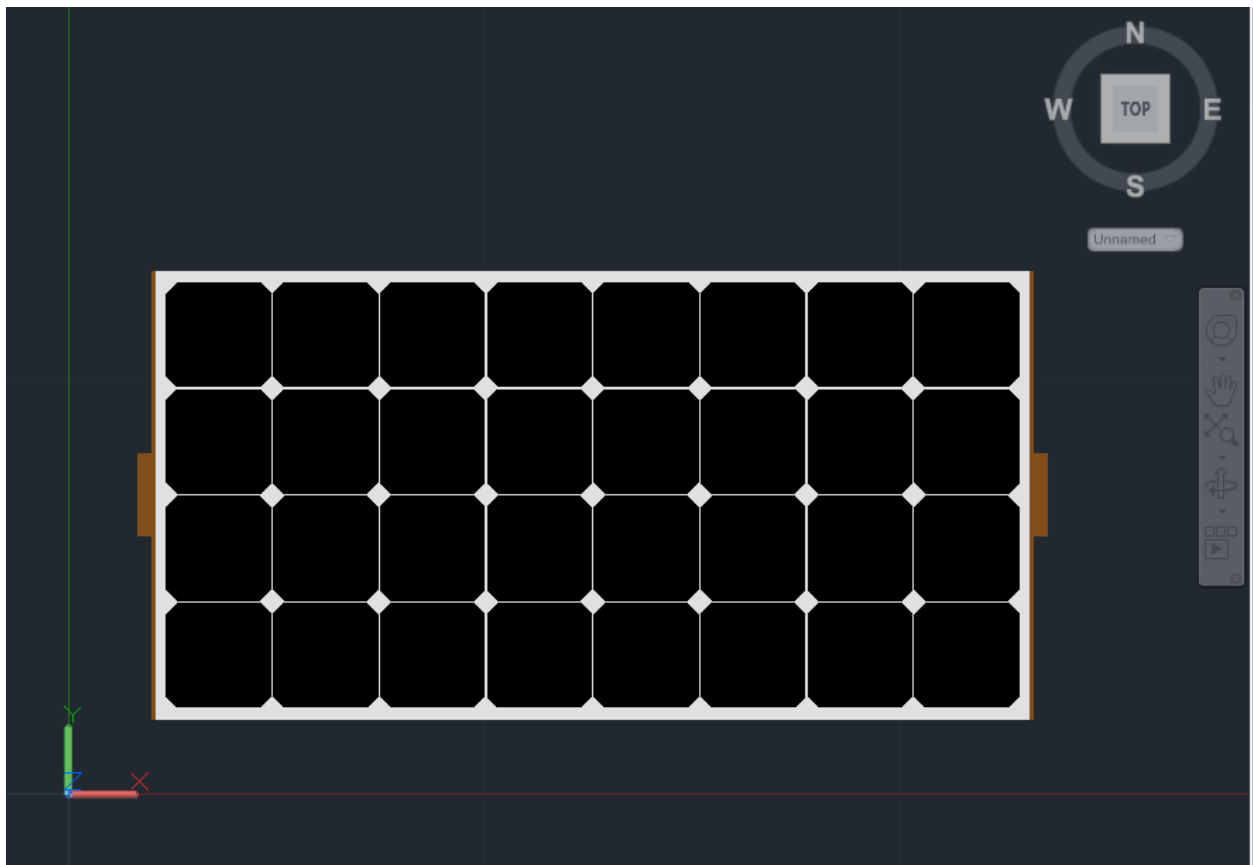


Fig. 4.6: Top view AutoCAD design of the framework

Figure 4.7 shows a picture of the final product of the solar tracker that comply with the design in AutoCAD.

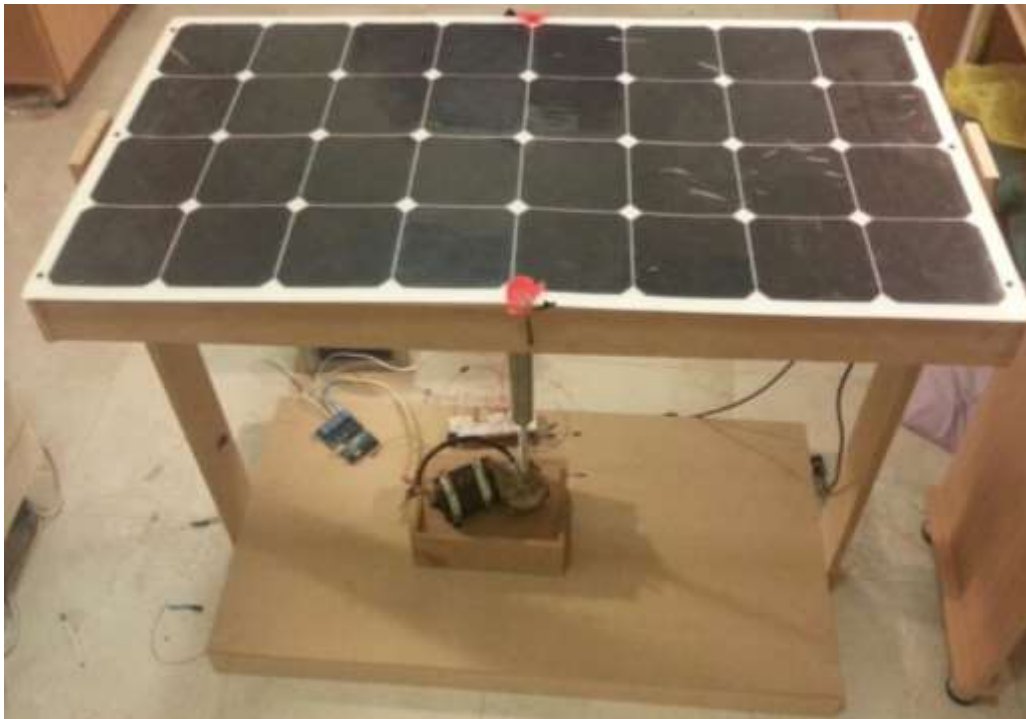


Fig. 4.7: Final product of the solar tracker

2.4 Design of Driving Motor

The driving motor is designed in such that is placed at the base of the tracker and connected with long screws to the center of the solar panel. In this way, the motor support the panel and hold it if there is a wind and it does not introduce weight on the structure of the panel. The motor is pulsed at different time intervals so it is at low speed. Its torque is 5.03 N-m and its speed is 1.26π rad/sec. It is connected with 12 volts battery, hence its current ratings can be calculated to be 1.67 Ampere. Figure 4.8 show the AutoCAD design of the base of the motor.

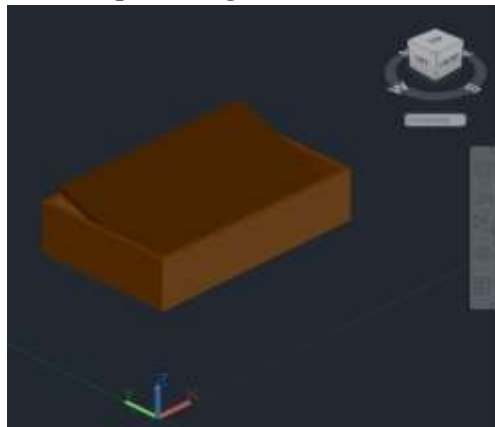


Fig. 4.8: AutoCAD design of the base supporting the motor

Figure 4.9 shows a picture of the constructed driving motor connected to a long screws and supporting the solar panel.



Fig. 4.9: Final product of the driving motor

4.5 Controlling Mechanism

In this project, we used a closed-loop tracker that is based of feedback control principles. In this closed-loop tracking control strategy, the search of the sun's position is implemented at any time of day; light sensors are used and positioned on the solar PV panel. If the sun is not facing the solar PV panel directly, there will be a difference in light intensity on one light sensor compared to another. This difference can be used to determine a direction which the tracker has to tilt in order to be facing the sun. Figure 4.10 shows a block diagram scheme of the closed-loop tracker.

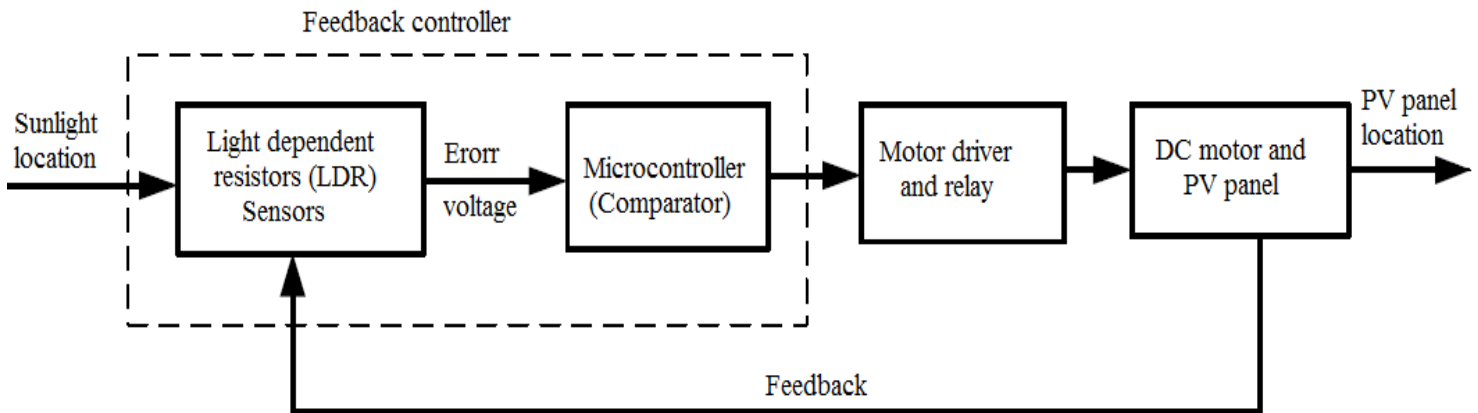


Fig. 4.10: block diagram scheme of the closed-loop tracker

In order to determine the sun's position, two similar light LDR sensors are mounted on the solar PV panel. They are located at the east and west to sense the light source intensity. There is an opaque objects at the two sensors which are to isolate the light from other orientations and to obtain a wide angle search and to determine the sun's position more quickly. Figure 4.11 shows a diagram of the solar cell and opaque objects besides the two sensors.

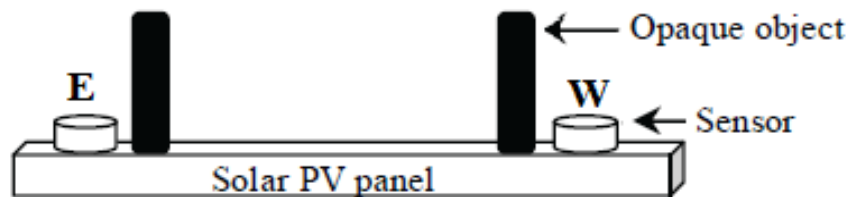


Fig. 4.11: Opaque objects besides the two sensors

There are two states considering in the scheme of the closed-loop tracker as follows. If the output signal values of the two sensors are equal, this also means that the output difference is zero and the motor's drive voltage is zero; then the solar tracker will be in the stable state. This means that the solar PV system has been tracked the current position of the sun as shown in Fig. 4.12. The other state of the closed-loop tracker occurs when the sun's position shifts as shown in Fig. 4.13. When the sensors receive the different light source intensities, the next step will be that the tracker makes a comparison for determining which sensor received more light. Finally, the tracker drives the motor to tilt the solar PV system which always perpendicular to the sun.

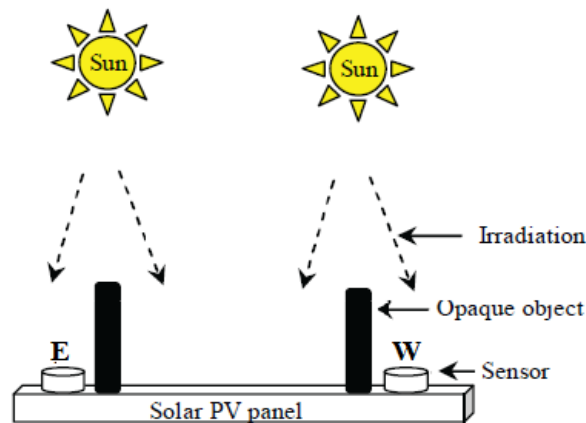


Fig. 4.12: two sensors receive equal light intensity

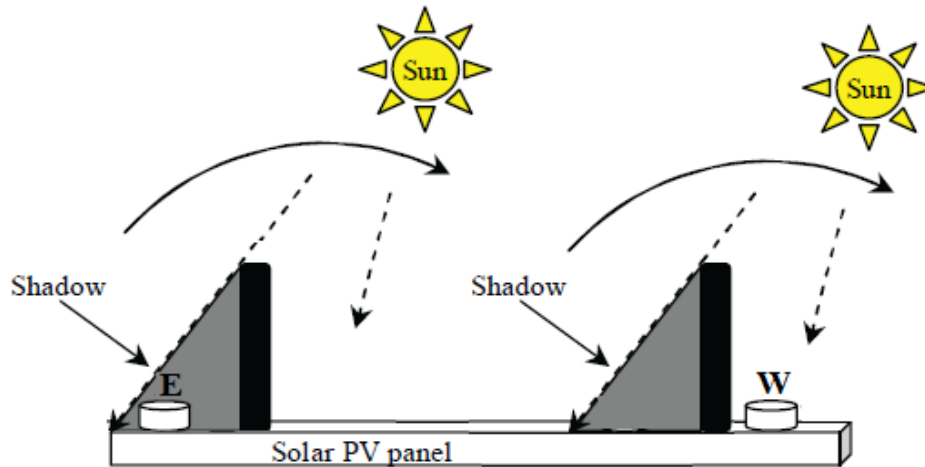


Fig. 4.13: One sensor receives more light intensity than the other

The used sensors are light dependent resistors (LDR) in the closed-loop tracker. The closed-loop tracker receives the signals which are the resistance values of two LDRs, R_E and R_W respectively. Then, it makes a comparison between R_E and R_W from time to time. The delay between each comparison is controlled by a potentiometer and the system will rotate if the difference between R_E and R_W is greater than a specified tolerance value, read from another potentiometer. Figure 4.14 shows a flow chart of the comparison process between the two sensors.

- If $|R_E - R_W| < \text{tolerance value}$
Then the solar PV panel will be kept in its position.
- If $|R_E - R_W| > \text{tolerance value}$, & $R_E < R_W$
Then there is much light in the west direction, and the system will rotate toward the west.
- If $|R_E - R_W| > \text{tolerance value}$, & $R_E > R_W$
Then there is much light in the east direction, and the system will rotate toward the east.

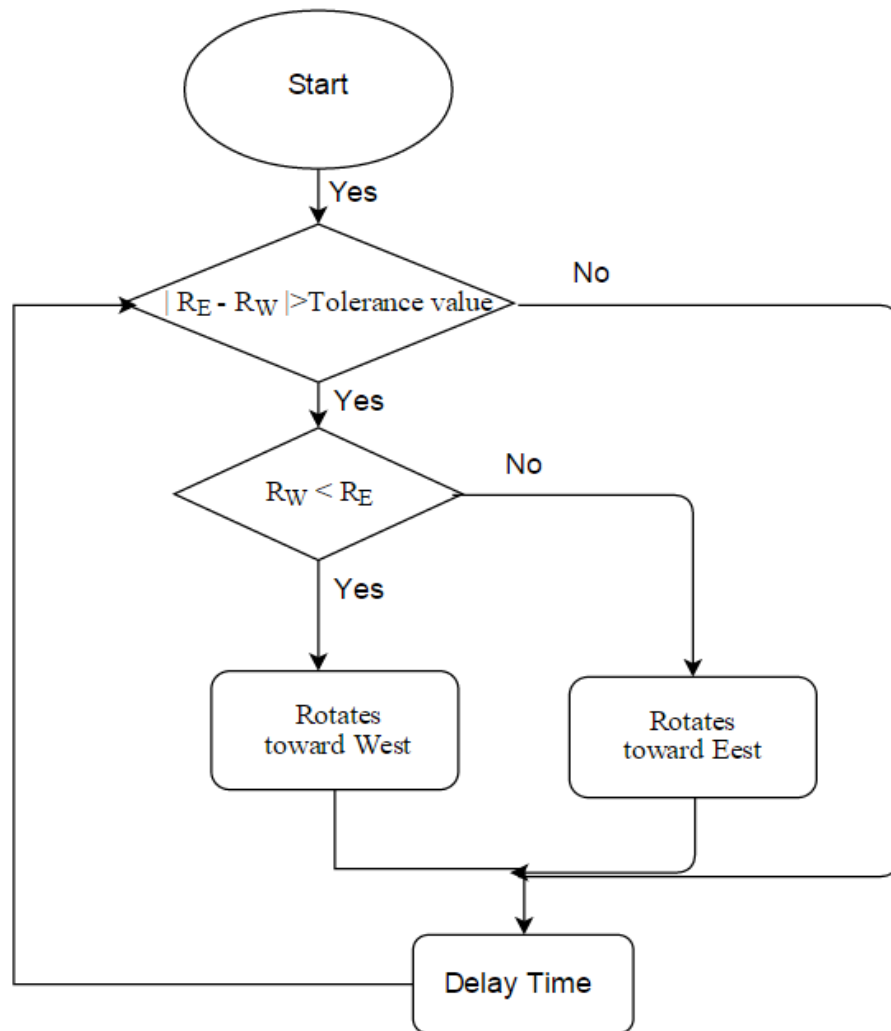


Fig. 4.14: Comparison process between the two sensors

Figure 4.15 show one LDR connected on top of the solar panel.



Fig. 4.15: One LDR connected on top of the solar panel

4.5 Interfacing & Circuit Connection

The electric circuit of this tracker is the circuit connecting the LDRs with the microcontroller, and the motor with relays and with the Arduino. Figure 4.16 shows the circuit diagram of the tracker system.

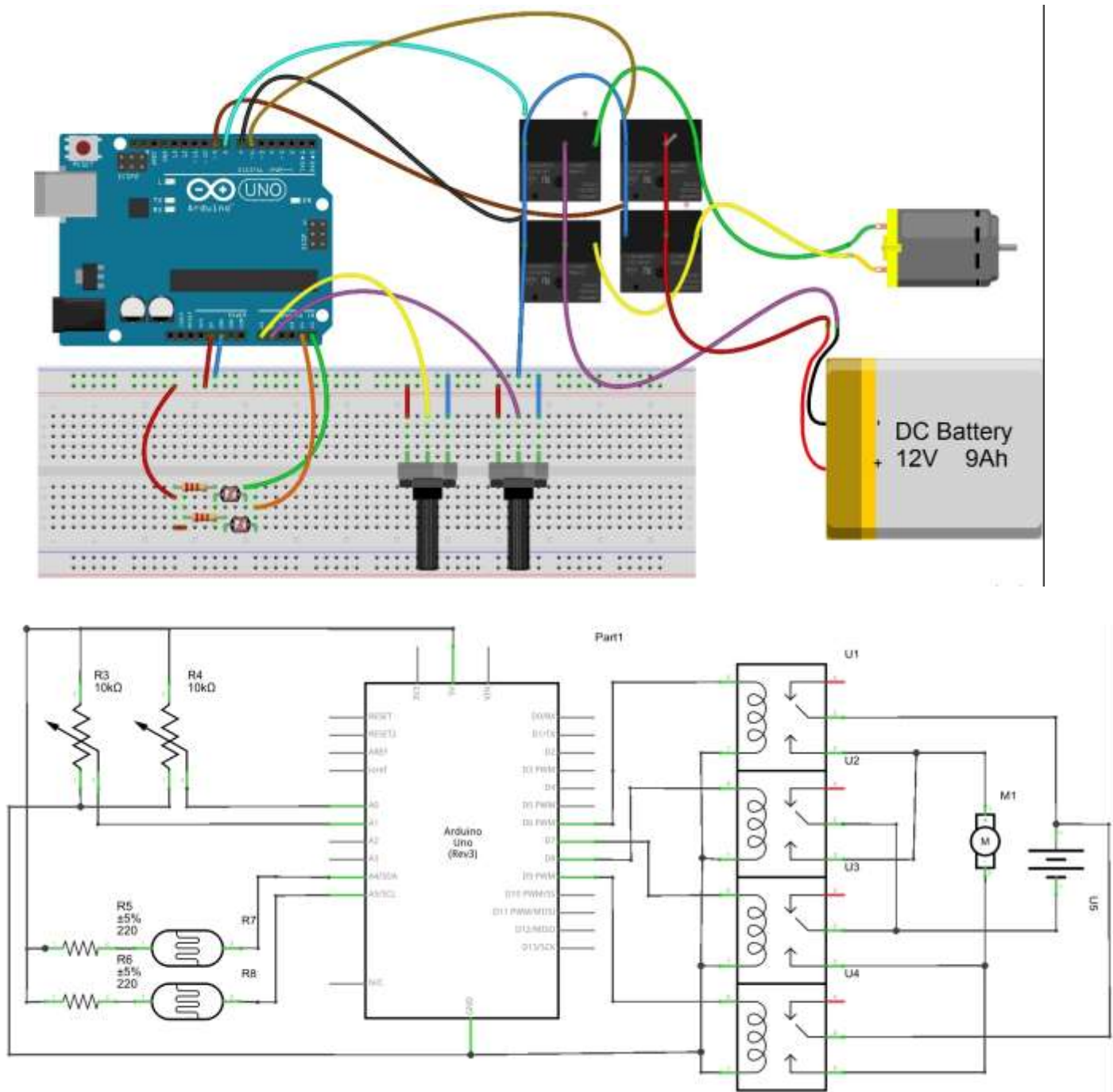


Fig. 4.16: Electric circuit diagram of the tracker system

CHAPTER 5

SOFTWARE

5.1 Arduino Programming Platform

To program the microcontroller, we used the C/C++ based Arduino programming language. Each software program written using the Arduino platform is called a sketch. Sketches do not include a `main()` function but instead use functions `setup()` and `loop()` working in conjunction to perform similar functionality. The `setup()` function will run only once at the beginning of execution. It is used to initialize serial port settings and configure input/output ports. Any local variables defined within `setup()` cannot be used in `loop()`. The `loop()` function will execute repeatedly until the Arduino is reset. The Arduino platform provides its own Integrated Development Environment (IDE) to develop software, upload programs, and communicate with Arduino hardware as shown in Fig. 5.1. The development environment contains a text editor for writing code, a console for error messages, and a toolbar. As well, there are a number of project templates and other features that can be accessed from the drop down menus.



Fig. 5.1: Arduino platform with Integrated Development Environment

The Serial Monitor records all activity over the serial port and displays it onscreen. This tool is particularly useful displaying the readings of light intensity of the two LDRs and for debugging the software because it can verify all input/output. By adjusting the baud rate so that it corresponds with the baud rate that was selected during initialization of the connection. Figure 5.2 shows the readings of light intensities and the two potentiometer displayed through serial monitor.

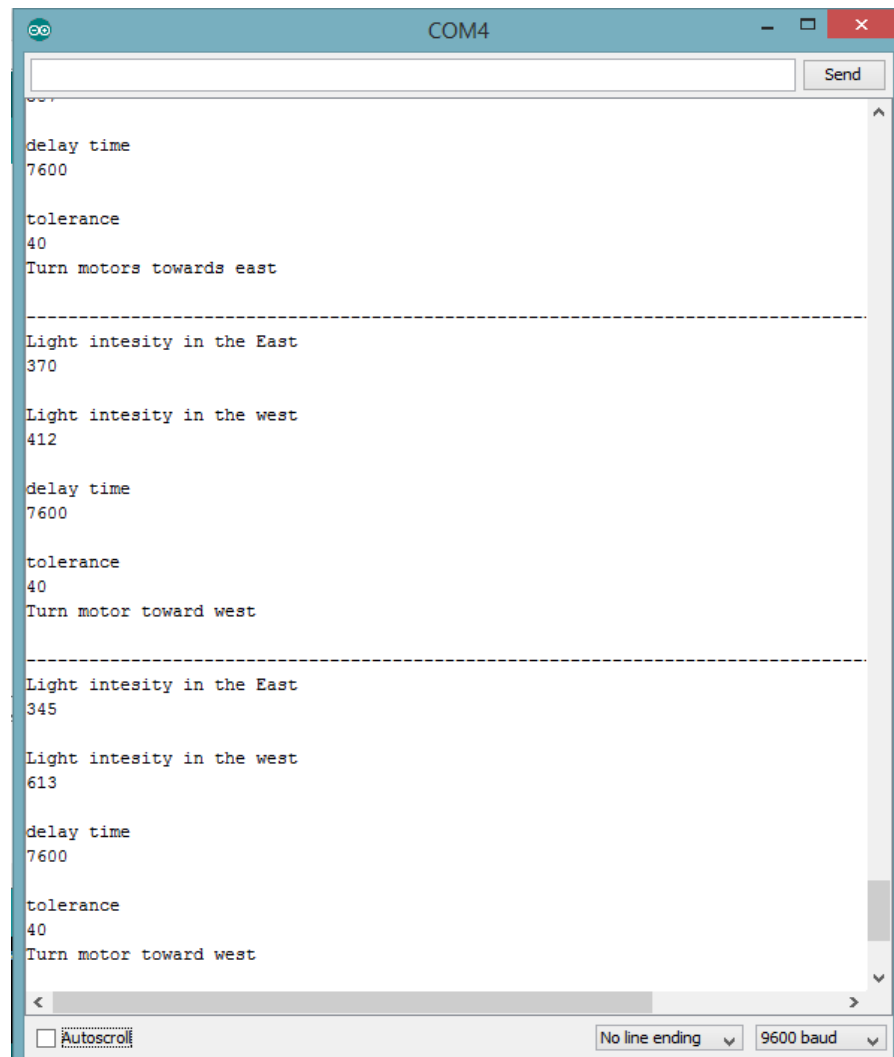


Fig. 5.2: Analog readings of the light displayed through serial monitor

5.2 Tracking Algorithm

To program the microcontroller, there must be a developed algorithm that visualize the whole process and aids in creating the programming code. Figure 5.3 shows the tracking algorithm we developed during programming the microcontroller.

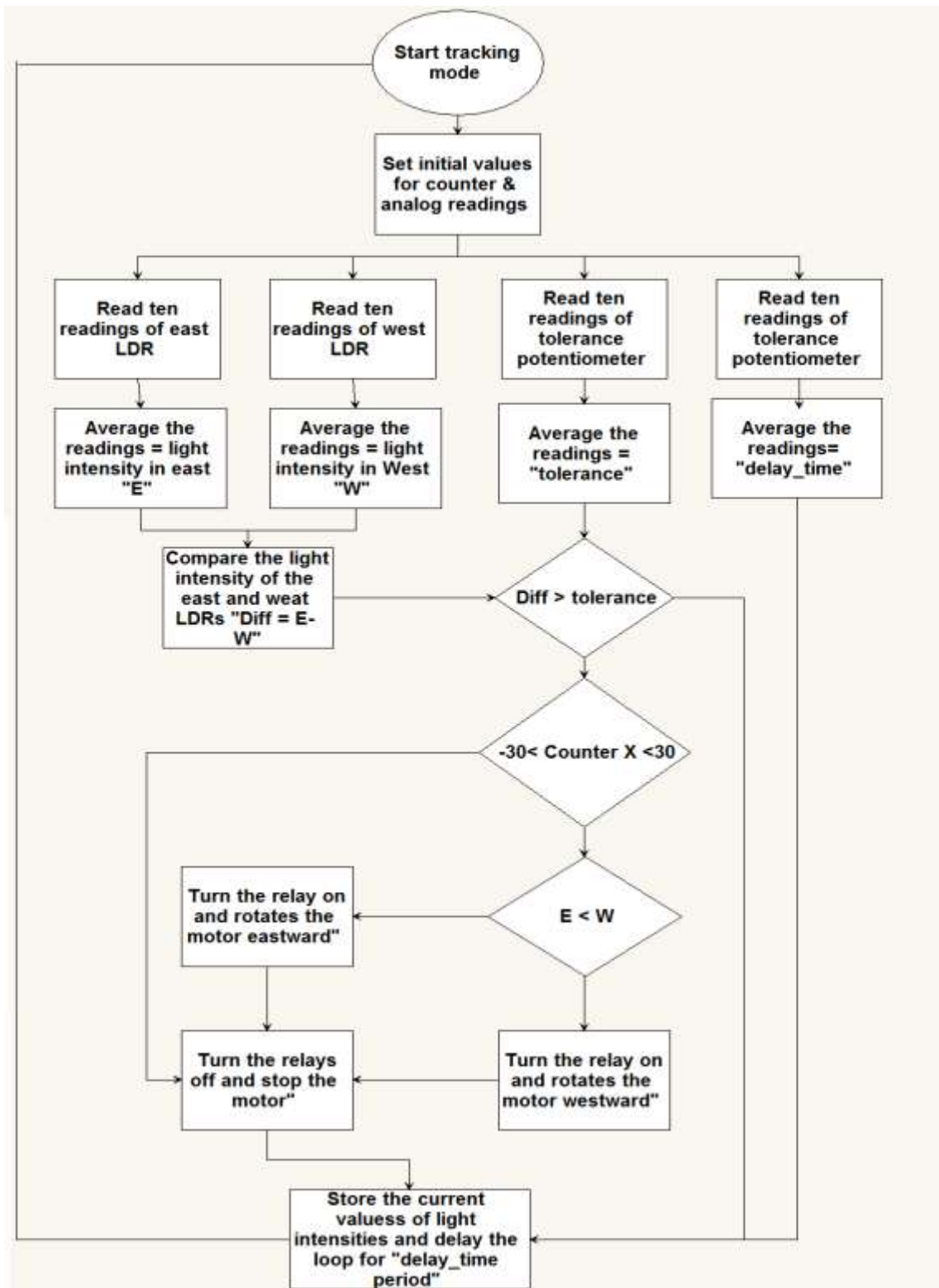


Fig. 5.3: Tracking algorithm used in programming the microcontroller

5.3 Software Implementation

By utilizing the developed tracking algorithm and using the Arduino Uno platform, we programmed the microcontroller. The code is in appendix A.

CHAPTER 6

EXPERIMENTAL VALIDATION

6.1 Overview

The solar tracker is kept under test for two days on 19th and 20th December. In the first day, the stationary solar panel (without tracking system) is tested to measure the current I and the short circuit current I_{sc} and to measure the voltage V on a load as well as the open circuit voltage V_{oc} . On the next day the same solar panel is tested but with solar tracker system in use to measure the voltage and current values. The measurement is done every hour from sunrise to sunset. The system is tested in Buraydah, Saudi Arabia, whose sun path diagram is shown in Fig. 6.1. As shown in the figure sun rise at 06:47 AM and set at 5.16 PM. The solar panel is placed with 120° from north so the azimuth angle comply with the sun path at these two days. In addition, the stationary solar panel is placed at 40° elevation angle from horizon, so it harness the maximum power at noon.

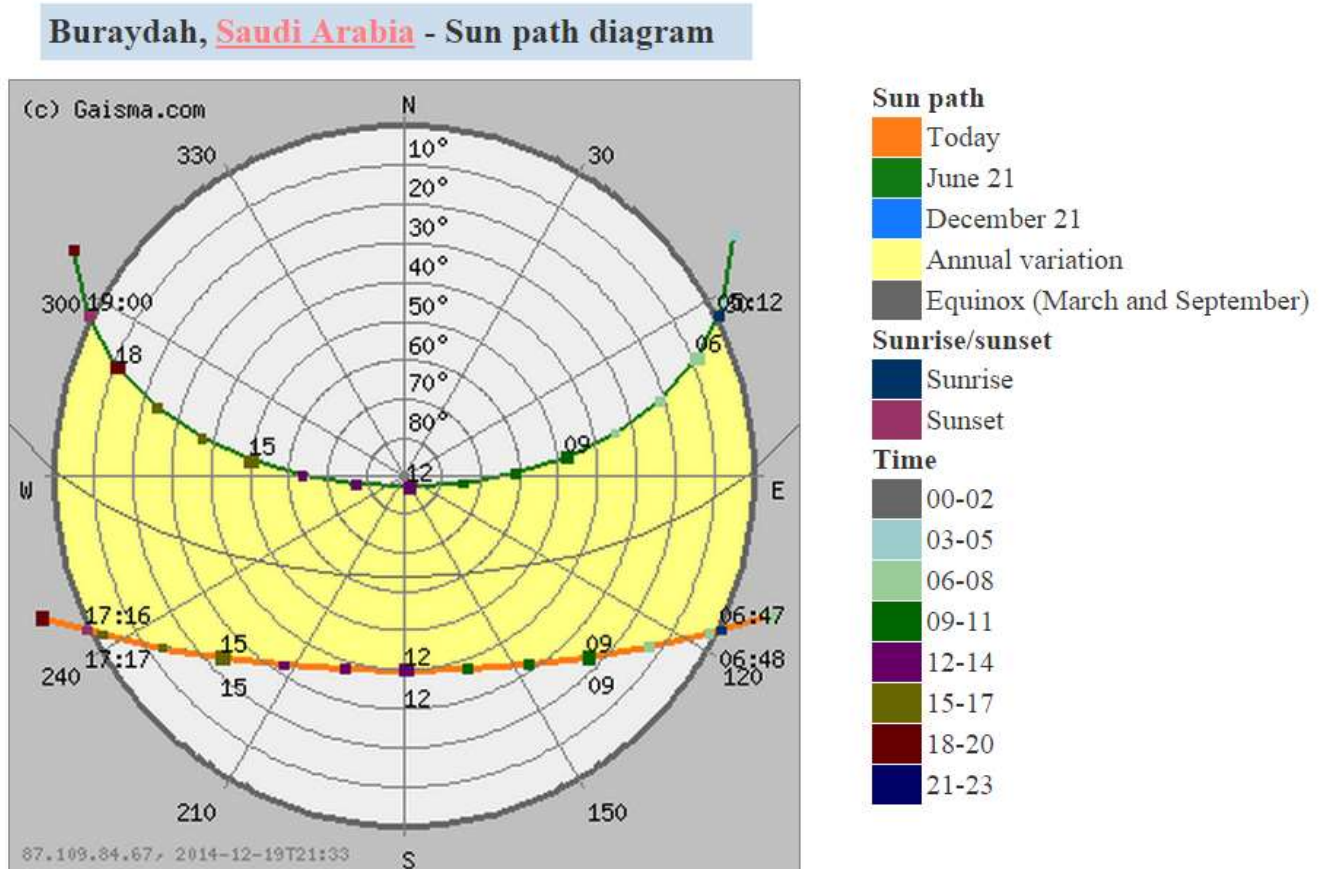


Fig. 6.1: Sun path diagram for Buraydah city on the test days [29]

Figure 6.2 shows the solar energy and surface meteorology data for Buraydah city. As shown, the insolation in December is very low, so this explains the reason for low output power of the cell compared to the maximum power in the product specification.

Buraydah, Saudi Arabia - Solar energy and surface meteorology

Variable	I	II	III	IV	V	VI	VII	VIII	IX	X	XI	XII
Insolation, kWh/m ² /day	3.22	4.40	5.36	6.18	6.93	7.65	7.47	7.09	6.34	5.24	3.63	2.96
Clearness, 0 - 1	0.50	0.57	0.59	0.60	0.63	0.68	0.67	0.67	0.66	0.64	0.53	0.49
Temperature, °C	12.30	14.73	19.07	24.85	30.78	33.56	34.98	35.20	32.92	27.93	20.60	14.67
Wind speed, m/s	4.87	5.25	5.32	5.03	4.95	5.53	5.80	5.45	4.85	4.77	5.19	4.87
Precipitation, mm	28	18	54	36	21	0	0	0	0	11	25	27
Wet days, d	2.0	1.5	2.2	1.6	0.5	0.0	0.0	0.0	0.0	0.3	1.0	1.3

Fig. 6.2: solar energy and surface meteorology data for Buraydah city [29]

Figure 6.3 shows some pictures of testing stationary and tracking solar panel.

**Fig. 6.3:** Pictures of testing the solar tracker

6.2 Experimental Results

The measurements of the solar tracker panel and the stationary solar panel are shown in Table 6.1 and Table 6.2, respectively/

Table 6.1: Measurements values of the solar tracker panel

Hour	Operating Voltage V (volts)	Operating Current I (Amp)	Open Circuit Voltage Voc (volts)	Short Circuit Current Isc (Amp)	Operating power P (Watts)
7:00 AM	1.15	0.94	15.6	1.12	1.081
8:00 AM	4.43	3.52	20.19	3.56	15.5936
9:00 AM	5.14	4.12	20.36	4.18	21.1768
10:00 AM	6.65	4.95	20.52	4.96	32.9175
11:00 AM	7.14	5.09	20.34	5.17	36.3426
12:00 PM	7.17	5.19	20.4	5.36	37.2123
1:00 PM	6.31	4.85	20.36	4.83	30.6035
2:00 PM	4.9	4.21	20.25	4.3	20.629
3:00 PM	3.29	3.15	20.08	3.18	10.3635
4:00 PM	2.91	2.67	20	2.86	7.7697

Table 6.2: Measurements values of the stationary solar panel

Hour	Operating Voltage V (volts)	Operating Current I (Amp)	Open Circuit Voltage Voc (volts)	Short Circuit Current Isc (Amp)	Operating power P (Watts)
7:00 AM	0.14	0.019	15.6	0.013	0.00266
8:00 AM	2.89	2.65	19	2.73	7.6585
9:00 AM	4.8	4.01	19.18	4.12	19.248
10:00 AM	5.41	4.15	20	4.2	22.4515
11:00 AM	6.73	4.87	20.3	4.9	32.7751
12:00 PM	5.56	4.68	20.4	4.71	26.0208
1:00 PM	4.12	3.76	19.7	3.8	15.4912
2:00 PM	1.83	2.18	19.6	2.2	3.9894
3:00 PM	0.9	0.5	18.1	0.53	0.45
4:00 PM	0.6	0.03	13.4	0.03	0.018
5:00 PM	0.002	0.001	6	0.001	0.000002

6.3 Results Analysis

The measurement values for the tracker solar panel in table 6.1 are graphed in Fig 6.4. From the graph, it is evident that the power increases smoothly from sunrise until it reaches its maximum values at noon, since the insolation value will be maximum. Then it will decrease smoothly afternoon.

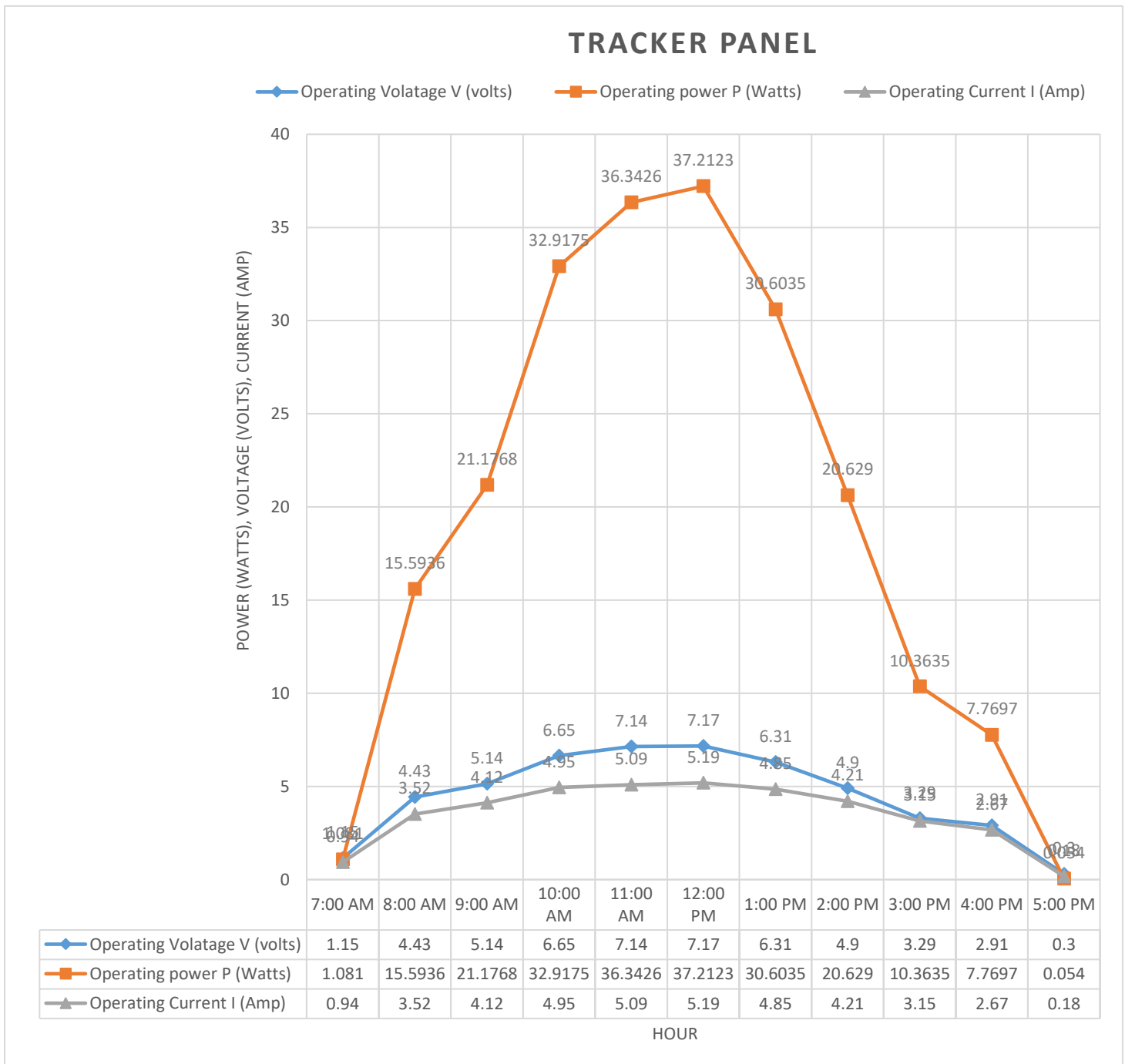


Fig. 6.4: Graph of voltage, current and power versus day's hours of tracker solar panel

Figure. 6.5 shows a graph of the measurement values in table 6.2 for the stationary solar panel. As the graph shows, the power maximum at noon since the panel is placed at 40° elevation angle in order to harness the maximum power available most at noon. The increase and decrease of the power is not smooth since the panel does not move at the change of radiation is great.

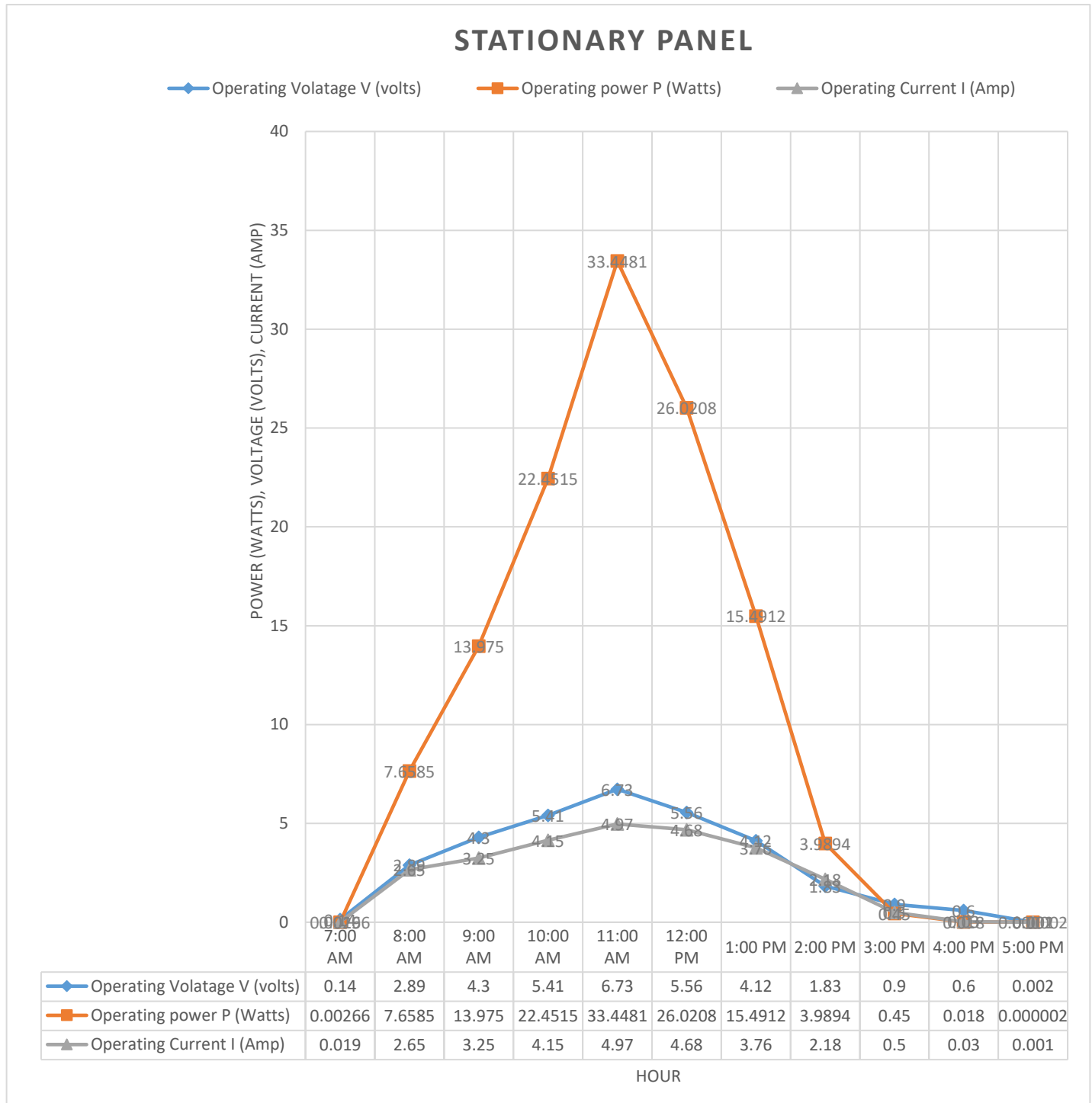


Fig. 6.5: Graph of voltage, current and power versus day's hours of stationary solar panel

Figure. 6.6 and figure. 6.7 show a graphical comparison of the voltage and current, respectively, produced by a tracker solar panel and a stationary solar panel.

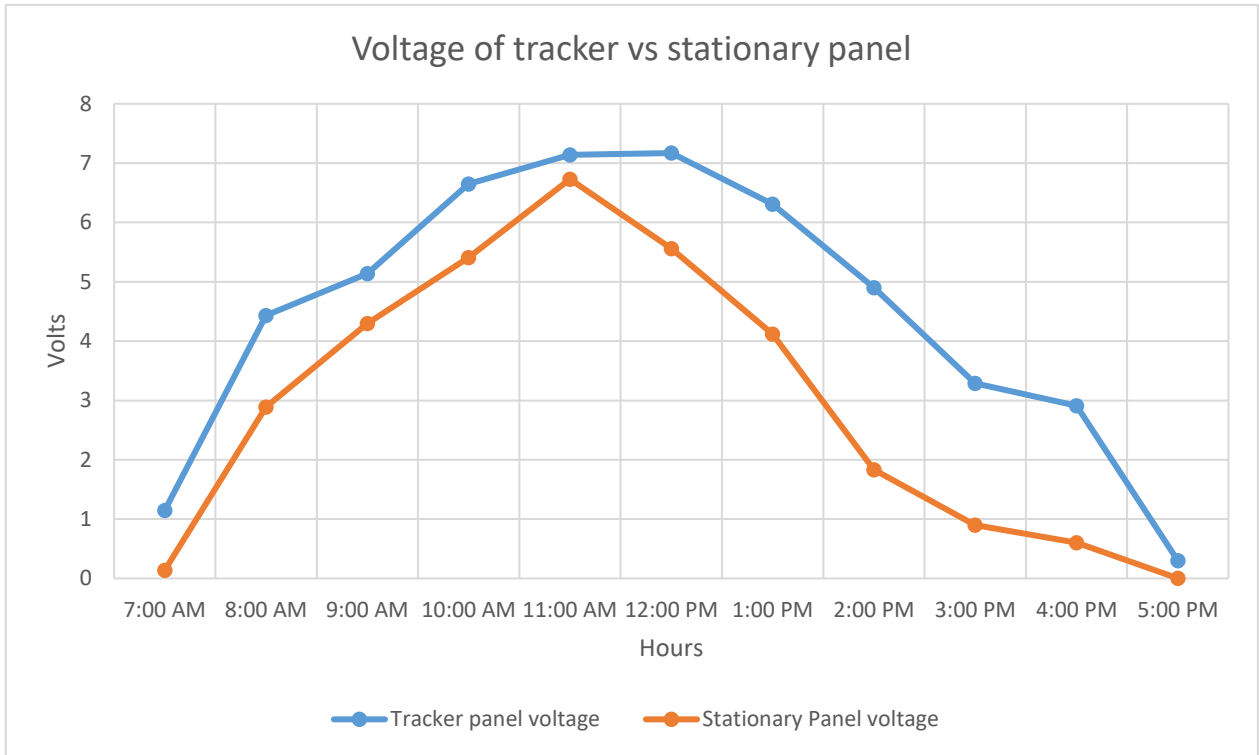


Fig. 6.6: Comparison of voltages produced by tracker and stationary solar panel

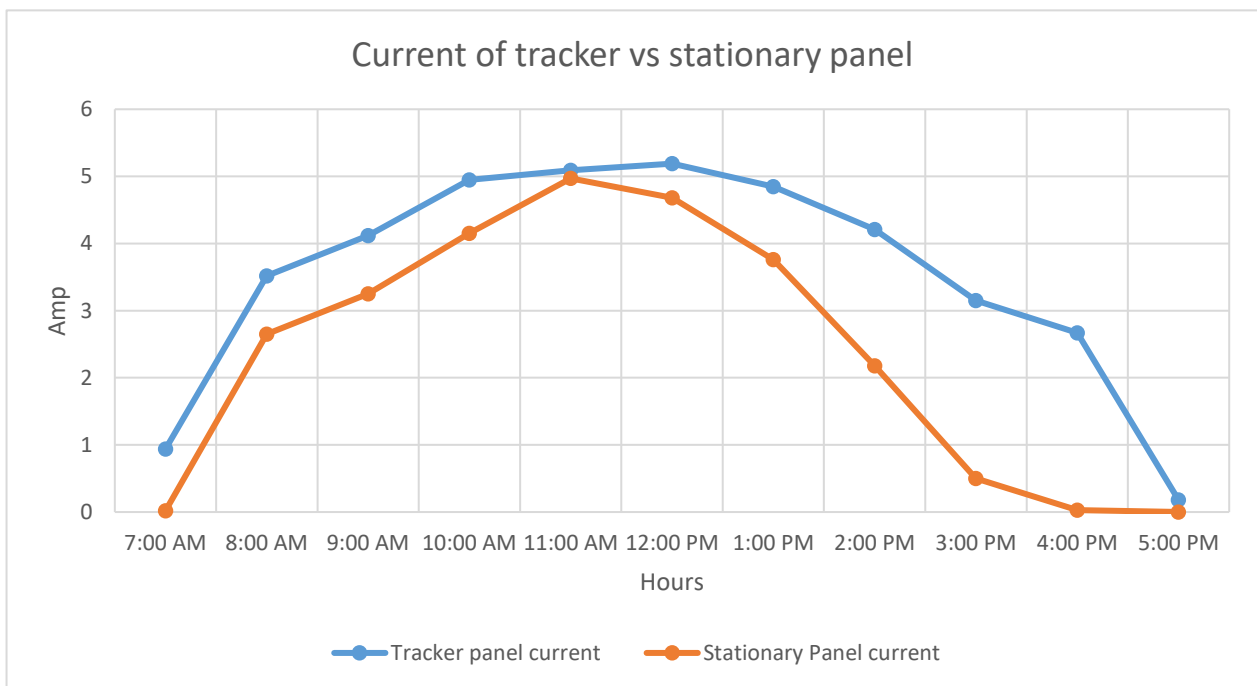


Fig. 6.7: Comparison of currents produced by tracker and stationary solar panel

Table 6.3 shows the difference in watts between the tracker panel and the stationary panel. For the whole day, the difference is approximately about 90 watt which is a considerable amount that when calculated for a long period exceed the capital cost of the tracker. From the table, the gain of output power of solar tracker is 57% higher than the stationary panel. Figure 6.8 shows a graphical comparison of the output power of the tracker and the stationary panel.

Table 6.3: Comparison of the output power of the tracker and stationary panel

Hour	Operating power of tracker panel (Watts)	Operating power of stationary panel (Watts)	Difference power of the tracker and stationary (Watts)
7:00 AM	1.081	0.00266	1.07834
8:00 AM	15.5936	7.6585	7.9351
9:00 AM	21.1768	13.975	7.2018
10:00 AM	32.9175	22.4515	10.466
11:00 AM	36.3426	33.4481	2.8945
12:00 PM	37.2123	26.0208	11.1915
1:00 PM	30.6035	15.4912	15.1123
2:00 PM	20.629	3.9894	16.6396
3:00 PM	10.3635	0.45	9.9135
4:00 PM	7.7697	0.018	7.7517
5:00 PM	0.054	0.000002	0.053998
Sum	213.7435	123.505162	90.238338

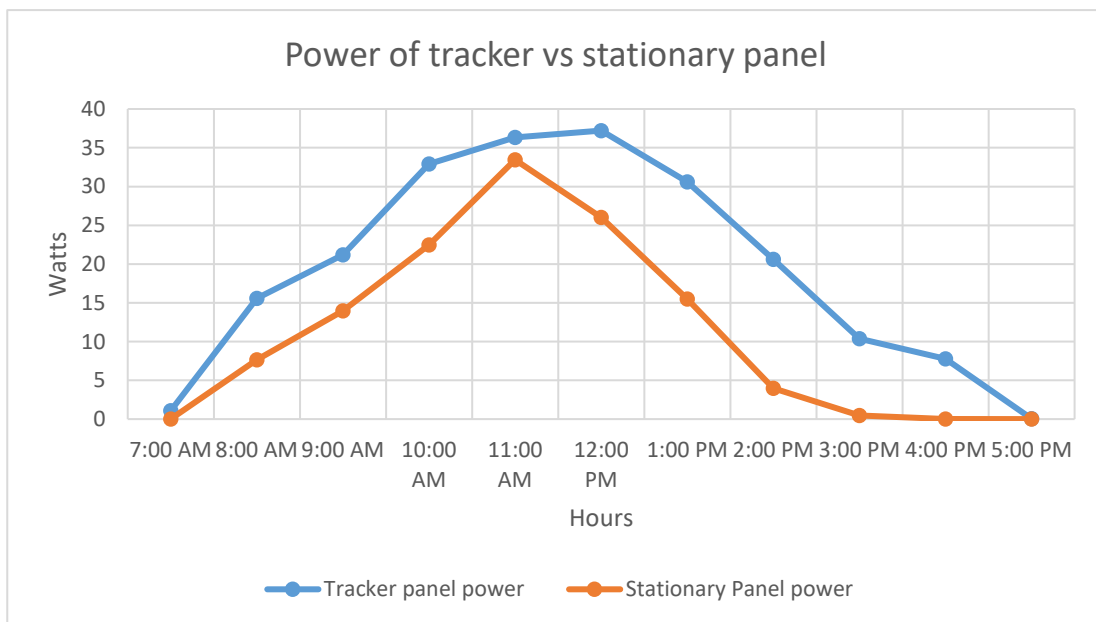


Fig. 6.8: Comparison of output power by tracker and stationary solar panel

CHAPTER 7

CONCLUSION

7.1 Conclusion

Solar energy conversion is favorable over other sources of energy due its characteristics of being clean, silent, and reliable, requires low maintenance cost and have low ecological impact. The most popular technology of solar conversion is the use of solar photovoltaics panels to convert solar energy to electricity. However; these photovoltaics panels have low efficiencies. One way of increasing their efficiencies is to maximize the capture of the rays of the sun upon the solar panels, which in turn increase the output power. A practical way of achieving this is by positioning the panels such that the rays of the sun fall perpendicularly on the solar panels by tracking the movement of the sun using an automatic solar tracker.

In our graduation project, we have designed and implemented one axis solar tracker that automatically controls solar tracking system to track solar PV panel according to the direction of beam propagation of solar radiation. This tracking system designed to automatically follow the sun and from the input received, it can actuate some mechanism to position the solar panel where it can receive maximum sunlight to produce more energy output then.

This system consists of basically a tow light sensors, one placed at the east and the other is placed at the west of the solar panel. These light sensors sense the light intensities in the east and the west directing, then send these data to a microcontroller. The microcontroller then compares readings from light sensors. If the difference of light intensities satisfied certain conditions such as it is greater than the tolerance value, the microcontroller instruct the motor to rotates toward the direction with high light intensity. To make the system more stable, the rotation of the motor is controlled with a counter, so it cannot rotates more than necessary. An electrical and mechanical design is first done before implementing the hardware and software.

Our system was tested in Buraydah on 18th and 19th of December. The test was to measure the operating current, voltage, and power as well as the open circuit voltage and short circuit current of the solar panel when the tracker system utilized and when the panel placed at a stationary state. A comparative analysis was performed between this solar tracker and a stationary PV panel. The results showed that gain of output power of PV in the solar tracker is 57% higher than the stationary PV panel.

7.2 Recommendation & Future Work

What is missed in our project is a reliable mechanical structure that can support large panels and afford extreme weather conditions. There are also some adjustment that can be done regarding to light sensors and the power supply.

The mechanical system in the future should be designed and analyzed properly to take account for large solar cells. This should be done by an experienced mechanical engineer who would be able to quickly determine errors and poor mechanical design. Aside from mechanical design adjustments, the motor system should be changed. There are three possible alternatives: a stepper motor, a geared motor, or hydraulics. Both the stepper motor and geared motor provide a significant decrease in the overall cost. Hydraulics might be better in power efficiency when moving large panel arrays, but it is uncertain if it would decrease cost. Finally, all the materials for the bearings and mechanical connections should be investigated by a materials engineer for outdoor use and altered accordingly.

There are a variety of solar sensors that were never tested or implemented. This primarily includes photodiodes and phototransistors, but also includes different photoresistor and photovoltaics. Further investigation in the dimensions and the spacing between the sensors should be conducted. Further investigation should also be done with the divider mount and the mount with a collimator for the sensors.

The battery could be utilized as a power source for the tracker and to store the power from the solar panel connected to the tracker. The use of the battery to store energy would require the design of a robust charging system. In addition this system should ideally integrate a maximum power point tracker (MPPT) and perhaps even incorporated into the same microcontroller being used for tracking to reduce the cost of adding this feature. MPPT is an efficient DC to DC convertor, which allows for the maximum power to be taken from the solar panel at any moment in time. Research should be focused on designing an MPPT and finding the most suitable battery.

The tracker can be improved further to be dual axis tracker instead of a single axis tracker to take account for seasonal changes. This can be done by incorporating another motor as well as two additional light sensors. The second motor can be installed at the base to rotate the whole structure whenever there is seasonal changes.

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Appendices

I. Programming Code

```
//set initial values
```

```
int x = 0;      // set initail zero value for the counter

int preE = 512; // set  initial value for light intensity in east direction

int preW = 512; // set  initial value for light intensity in west direction

int preptime = 512; // set  initial value for delay time

int pretol= 512; // set  initial value for tolerance value
```

```
void setup()
```

```
{
```

```
    Serial.begin(9600); // set the baud rate for the serial monitor
```

```
// first set the relays off
```

```
    pinMode(7, OUTPUT);
```

```
    pinMode(8, OUTPUT);
```

```
    digitalWrite(7,HIGH);
```

```
    digitalWrite(8,HIGH);
```

```
    // began the loop
```

```
}
```

```
void loop()
```

```
{
```

```
    // Read the first reading
```

```
    //read the east LDR
```

```
        int E1 = analogRead(5);
```

```
    //read the west LDR
```

```
        int W1 = analogRead(4);
```

```
    // read the delay time potentiometer
```

```
        int dtime1 = analogRead(1);
```

```
//read the tolerance potentiometer

    int tol1 = analogRead(0);

// Check if the readings reasonable, otherwise set them to previous ones.

    if (E1 < 10 || E1 > 1000 )
    {
        E1 = preE ;
    }

    if (W1 < 10 || W1 > 1000 )
    {
        W1 = preW ;
    }

    if (dtime1 < 10 || dtime1 > 1000 )
    {
        dtime1 = pretime ;
    }

    if (tol1 < 10 || tol1 > 1000)
    {
        tol1 = pretol ;
    }

    delay(100);

// delay for one 0.1 second and take the next reading
```

```
int E2= analogRead(5);
int W2= analogRead(4);
int dtime2= analogRead(1);
int tol2= analogRead(0);

if (E2 < 10 || E2 > 1000 )
{
    E2= preE ;
```

```

}
if (W2 < 10 || W2 > 1000 )
{
    W2= preW ;
}
if (dtime2 < 10 || dtime2 > 1000 )
{
    dtime2= pretime ;
}
if (tol2 < 10 || tol2 > 1000 )
{
    tol2= pretol ;
}
delay(100);

```

```
// delay for one 0.1 second and take the third reading
```

```

int E3= analogRead(5);
int W3= analogRead(4);
int dtime3= analogRead(1);
int tol3= analogRead(0);
if (E3 < 10 || E3 > 1000 )
{
    E3= preE ;
}
if (W3 < 10 || W3 > 1000 )
{
    W3= preW ;
}
if (dtime3 < 10 || dtime3 > 1000 )

```

```

{
    dtime3= predtime ;
}
if (tol3 < 10 || tol3 > 1000 )
{
    tol3= pretol ;
}

```

```

int E4 = analogRead(5);
int W4 = analogRead(4);
int dtime4= analogRead(1);
int tol4= analogRead(0);
if (E4 < 10 || E4 > 1000 )
{
    E4= preE ;
}
if (W4 < 10 || W4 > 1000 )
{
    W4= preW ;
}
if (dtime4 < 10 || dtime4 > 1000 )
{
    dtime4= predtime ;
}
if (tol4 < 10 || tol4 > 1000 )
{
    tol4= pretol ;
}

```

```

delay(100);
int E5 = analogRead(5);
int W5 = analogRead(4);
int dtime5= analogRead(1);
int tol5= analogRead(0);
if (E5 < 10 || E5 > 1000 )
{
    E5= preE ;
}
if (W5 < 10 || W5 > 1000 )
{
    W5= preW ;
}
if (dtime5 < 10 || dtime5 > 1000 )
{
    dtime5= pretime ;
}
if (tol5 < 10 || tol5 > 1000 )
{
    tol5= pretol ;
}
delay(100);
int E6 = analogRead(5);
int W6 = analogRead(4);
int dtime6= analogRead(1);
int tol6= analogRead(0);
if (E6 < 10 || E6 > 1000 )
{

```

```
E6= preE ;  
}  
if (W6 < 10 || W6 > 1000 )  
{  
    W6= preW ;  
}  
if (dtime6 < 10 || dtime6 > 1000 )  
{  
    dtime6= pretime ;  
}  
if (tol6 < 10 || tol6 > 1000 )  
{  
    tol6= pretol ;  
}  
delay(100);  
int E7 = analogRead(5);  
int W7 = analogRead(4);  
int dtime7= analogRead(1);  
int tol7= analogRead(0);  
if (E7 < 10 || E7 > 1000 )  
{  
    E7= preE ;  
}  
if (W7 < 10 || W7 > 1000 )  
{  
    W7= preW ;  
}  
if (dtime7 < 10 || dtime7 > 1000 )
```

```

{
    dtime7= predtime ;
}
if (tol7 < 10 || tol7 > 1000 )
{
    tol7= pretol ;
}
delay(100);
int E8 = analogRead(5);
int W8 = analogRead(4);
int dtime8= analogRead(1);
int tol8= analogRead(0);
if (E8 < 10 || E8 > 1000 )
{
    E8= preE ;
}
if (W8 < 10 || W8 > 1000 )
{
    W8= preW ;
}
if (dtime8 < 10 || dtime8 > 1000 )
{
    dtime8= predtime ;
}
if (tol8 < 10 || tol8 > 1000 )
{
    tol8= pretol ;
}

```

```

delay(100);
int E9 = analogRead(5);
int W9 = analogRead(4);
int dtime9= analogRead(1);
int tol9= analogRead(0);
if (E9 < 10 || E9 > 1000 )
{
    E9= preE ;
}
if (W9 < 10 || W9 > 1000 )
{
    W9= preW ;
}
if (dtime9 < 10 || dtime9 > 1000 )
{
    dtime9= pretime ;
}
if (tol9 < 10 || tol9 > 1000 )
{
    tol9= pretol ;
}
delay(100);
int E10 = analogRead(5);
int W10 = analogRead(4);
int dtime10= analogRead(1);
int tol10= analogRead(0);
if (E10 < 10 || E10 > 1000 )
{

```



```

    E10= preE ;
}
if (W10 < 10 || W10 > 1000 )
{
    W10= preW ;
}
if (dtime10 < 10 || dtime10 > 1000 )
{
    dtime10= predtime ;
}
if (tol10 < 10 || tol10 > 1000 )
{
    tol10= pretol ;
}

//Average the ten readings

int E=(E1+E2+E3+E4+E5+E6+E7+E8+E9+E10)/10; //light intensity in east
int W =(W1+W2+W3+W4+W5+W6+W7+W8+W9+W10)/10; // light intensity in west
int
dtime=(dtime1+dtime2+dtime3+dtime4+dtime5+dtime6+dtime7+dtime8+dtime9+dtime10)/10;

int tol = (tol1+tol2+tol3+tol4+tol5+tol6+tol7+tol8+tol9+tol10)/10;// tolerance value
int delaytime = (dtime *10);// increase the delay by magnitude of 10
int tolerance = tol / 40;

// display the readings through serial monitor

Serial.println("Light intesity in the East");
Serial.println(E);
Serial.println("      ");
Serial.println("Light intesity in the west");
Serial.println(W);

```

```
Serial.println(" ");
Serial.println("delay time");
Serial.println(delaytime);
Serial.println(" ");
Serial.println("tolerance");
Serial.println(tolerance);
```

```
//check the differenc and open the coresponding relays
```

```
// check the diffirence of light intensity in east and west
```

```
int diff = E - W;
```

```
// check if the diffirence is in the tolerance else rotates the motor
```

```
if (-1*tolerance > diff || diff > tolerance)
{
if (E > W)// check if light intensity in east is greater than in west
{
if ( x <30)// check counter to ensure the motor does not run infinitely
{
Serial.println("Turn motors towards east");
digitalWrite(7, LOW);// Turn ON Relay 1
delay(2000);
digitalWrite(7,HIGH); //Turn OFF Relay 1
++x;
}
}
}
```

```
//check if light intensity in west is greater than in east
```

```
else if (W > E)
{
if (x > -30)
{
```

```

Serial.println("Turn motor toward west");
digitalWrite(8, LOW); // Turn ON Relay 4
digitalWrite(9, LOW);
delay(2000);
digitalWrite(8, HIGH); // Turn OFF Relay 4
digitalWrite(9, HIGH);
--x; // decrease counter
}
}
}
//store the values for the next loop
preE = E;
preW = W;
pretol= tol;
predtime = dtime;
Serial.println(" ");
Serial.println("----- ");
// deley the next loop
delay(delaytime);
}

```

II. Arduino Uno Pins

The power pins are as follows:

- **VIN:** The input voltage to the Arduino board when it's using an external power source (as opposed to 5 volts from the USB connection or other regulated power source). You can supply voltage through this pin, or, if supplying voltage via the power jack, access it through this pin.
- **5V:** The regulated power supply used to power the microcontroller and other components on the board. This can come either from VIN via an on-board regulator, or be supplied by USB or another regulated 5V supply.
- **3V3.** A 3.3 volt supply generated by the on-board regulator. Maximum current draw is 50 mA.
- **GND:** Ground pins.
-

Each of the 14 digital pins on the Uno can be used as an input or output, using [pinMode\(\)](#), [digitalWrite\(\)](#), an [digitalRead\(\)](#) functions. They operate at 5 volts. Each pin can provide or receive a maximum of 40 mA and has an internal pull-up resistor (disconnected by default) of 20-50 k Ohms. In addition, some pins have specialized functions:

- **Serial: 0 (RX) and 1 (TX).** Used to receive (RX) and transmit (TX) TTL serial data. These pins are connected to the corresponding pins of the ATmega8U2 USB-to-TTL Serial chip .
- **External Interrupts: 2 and 3.** These pins can be configured to trigger an interrupt on a low value, a rising or falling edge, or a change in value.
- **PWM: 3, 5, 6, 9, 10, and 11.** Provide 8-bit PWM output with the [analogWrite\(\)](#) function.
- **SPI: 10 (SS), 11 (MOSI), 12 (MISO), 13 (SCK).** These pins support SPI communication, which, although provided by the underlying hardware, is not currently included in the Arduino language.
- **LED: 13.** There is a built-in LED connected to digital pin 13. When the pin is HIGH value, the LED is on, when the pin is LOW, it's off.

The Uno has 6 analog inputs, each of which provide 10 bits of resolution (i.e. 1024 different values). By default they measure from ground to 5 volts, though it is possible to change the upper end of their range using the AREF pin and the `analogReference()` function. Additionally, some pins have specialized functionality:

- **I2C: 4 (SDA) and 5 (SCL).** Support I2C (TWI) communication using the [Wire library](#).

There are a couple of other pins on the board:

- **AREF.** Reference voltage for the analog inputs. Used with `analogReference()`.
- **Reset.** Bring this line LOW to reset the microcontroller. Typically used to add a reset button to shields which block the one on the board.

